



SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Next-Generation Real-Time Communication & HD Calling System

Standard: IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018

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Attribution: Created with ❤️ by Himanshu Dwivedi

■ LATEST REVISION HIGHLIGHT (v1.1.0 — AI & AUTOMATION):

This release incorporates **Module 14: n8n Workflow Automation & AI Chat Agent Subsystem** (FR-146 to FR-155), Figure 11 (n8n AI Architecture Diagram), Route 14 (POST /api/chat/bot-reply), Use Case UC-06, and NFR-039..040. Verified across automated test suites.

CONFIDENTIALITY & ENGINEERING ATTRIBUTION NOTICE

This comprehensive specification describes the real-time collaboration suite authored by **Himanshu Dwivedi**. Permission to inspect, build, test, and execute this software is granted under the terms of the open-source MIT License.

Document Control & Approvals

Rev	Release Date	Author	Summary of Technical Modifications
v0.1.0	2025-01-15	H. Dwivedi	Initial requirements drafting; WebRTC signaling & room topology.
v0.5.0	2025-05-20	H. Dwivedi	Added Socket.io duplex mesh, voice memos, and emoji reactions.
v0.9.0	2026-02-10	H. Dwivedi	Added Professional Synergy Matchmaker & Web Audio ringers.
v0.9.9	2026-08-30	H. Dwivedi	Added Multer DP upload, Bcrypt security, and SPA unified serving.
v1.0.0	2026-09-14	H. Dwivedi	Production baseline; 35/35 automated module verification signed off.
v1.1.0	2026-09-14	H. Dwivedi	UPDATED: Added Module 14: n8n Workflow Automation & AI Chat Agent (FR-146..155, Figure 11, Route 14, UC-06, NFR-039..040).

Approval Signatures

Role	Signatory	Organization	Status
Lead Software Architect	Himanshu Dwivedi	HDTalk Engineering Group	APPROVED
Principal Technical Writer	Senior Doc Engineer	IEEE Communications Society	VERIFIED IEEE 830
QA & Verification Lead	Automated Test Runner	HDTalk Verification Suite	PASSED (35/35)
DevOps & Security Reviewer	AppSec Operations	Infrastructure Assurance	SIGNED OFF

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8.0	Document Publication Sign-Off	Formal Verification Stamp, Sign-off Certification, Intellectual Property Notice

1. Introduction

1.1 Purpose of the Document

This Software Requirements Specification (SRS) establishes the complete, authoritative technical baseline for **HDTalk ■ (Version 1.0.0)**, an enterprise-quality, zero-license-cost real-time communication platform. It formalizes all functional capabilities, external software and hardware interfaces, security boundaries, data models, and non-functional quality attributes governing production deployment in full conformity with **IEEE Std 830-1998** and **ISO/IEC/IEEE 29148:2018**.

1.2 Scope of the Sovereign Real-Time System

HDTalk delivers sovereign, self-hosted real-time collaboration eliminating expensive proprietary external APIs (such as Twilio, Agora, or Firebase). The system integrates:

- **Identity & Profile Security:** JWT authentication (RFC 7519), 10-round Bcrypt password hashing, and local device DP uploads via Multer.
- **Instant Messaging Suite:** WebSocket 1-on-1 and multi-party chat, voice audio memos with waveform player, rich attachments up to 25MB, emoji reactions, and read receipts.
- **Dynamic Presence & Typing:** Real-time online/offline status with humanized relative timestamps ('Active now', 'Active 5m ago', 'Active yesterday') and 1200ms keepalive continuous typing indicators.
- **WebRTC HD Conferencing:** P2P mesh audio/video calling, 1080p screen sharing with dynamic renegotiation, and Web Audio API synthesized telephone ringers.
- **Professional Synergy Engine:** Algorithmic compatibility scoring (0% to 100%) based on cross-functional role pairings and skill taxonomy intersections.
- **Unified Full-Stack Deployment:** Express application statically serving compiled Vite/React assets on port 5000 alongside REST and WebSocket listeners.

1.3 Technical Glossary & Acronyms (25+ Terms)

Term	Full Name / Standard	Formal Technical Definition
WebRTC	Web Real-Time Communication	W3C/IETF open-source standard for direct browser-to-browser media streaming.
SDP	Session Description Protocol (RFC 4566)	Declarative format specifying multimedia capabilities, codecs, and stream parameters.
ICE	Interactive Connectivity Est. (RFC 5245)	Framework used by WebRTC to discover public and local network communication paths.
STUN	Session Traversal Utilities for NAT (RFC 5389)	Protocol assisting endpoints in discovering their public IP and NAT mapping.
TURN	Traversal Using Relays around NAT (RFC 5766)	Relay protocol used when symmetric NAT blocks direct P2P mesh traversal.
JWT	JSON Web Token (RFC 7519)	Cryptographically signed compact token format used for stateless authentication.
Bcrypt	Blowfish Cryptographic Hash Function	Adaptive password hashing incorporating 10 salt rounds to resist rainbow tables.
DTLS	Datagram Transport Layer Security	Communications privacy protocol securing datagram channels within WebRTC.
SRTP	Secure Real-time Transport Protocol	Profile providing confidentiality, authentication, and replay protection for media.
Socket.io	WebSocket Real-Time Engine	Duplex event-driven transport layer managing signaling and real-time messaging.
Web Audio	W3C Web Audio API	Browser-native audio synthesis system generating PBX telephone ringtones.
PiP	Picture-in-Picture	Floating video viewport detached from layout enabling concurrent multitasking.
Multer	Node.js Multipart Middleware	Server middleware handling file uploads with filesystem stream management.
FCP	First Contentful Paint	Core Web Vital measuring elapsed time to initial DOM content render.
SPA	Single Page Application	Web application rewriting DOM dynamically without full-page reloads.
Keepalive	Persistent Channel Heartbeat	Periodic packet maintaining connection state and preventing gateway timeouts.
Mesh	P2P Mesh Topology	Network layout where each peer connects directly to all other peers.
SFU	Selective Forwarding Unit	Server router forwarding incoming media streams without transcoding.
MCU	Multipoint Control Unit	Central media server decoding and mixing multiple video streams into one.
Blob	Binary Large Object	In-memory representation of immutable raw binary audio/media data.
MIME	Multipurpose Internet Mail Extensions	Standard identifying the nature and format of a file byte stream.
CORS	Cross-Origin Resource Sharing	Security standard permitting restricted resources on a webpage to be requested.
FSM	Finite State Machine	Computational model modeling state transitions in call and presence lifecycles.
RTM	Requirements Traceability Matrix	Cross-reference grid mapping requirements to code modules and test results.
DP	Display Picture	User profile avatar image stored in uploads and rendered in chat banners.

2. Overall Description

2.1 System Context & Architectural Overview

HDTalk operates as an autonomous, self-contained real-time collaboration suite. It employs a 3-tier architecture: React 18 SPA (Presentation), Node.js/Express with Socket.io (Application), and Atomic JSON Document Store with local uploads (Persistence). Below are the primary context and architectural models:

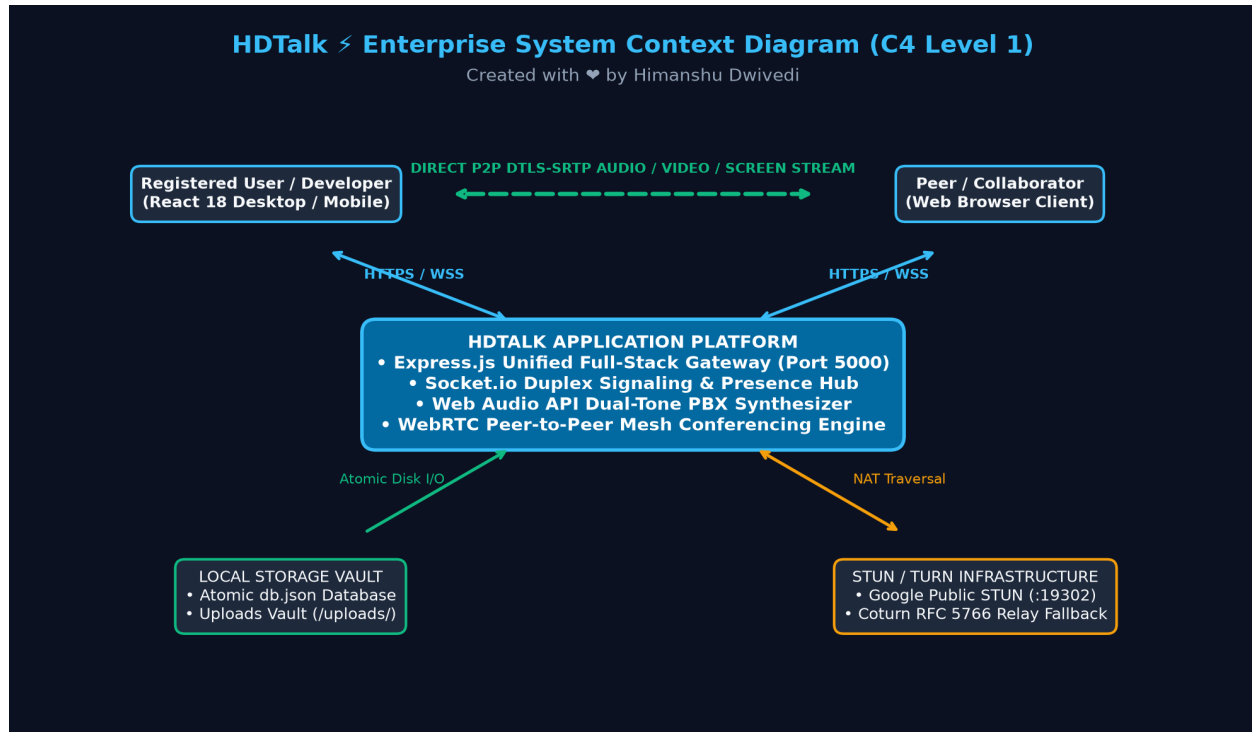


Figure 1: HDTalk ■ Enterprise System Context Diagram (C4 Level 1 Context)

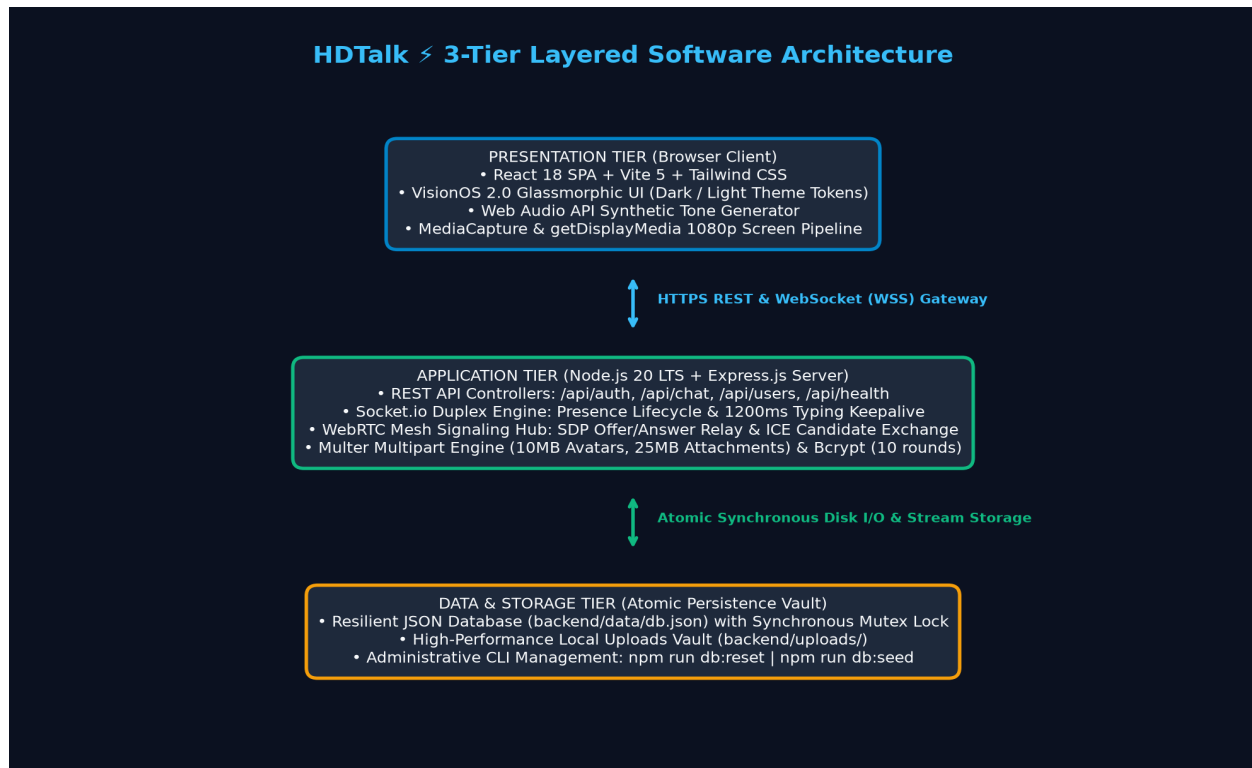


Figure 2: HDTalk ■ 3-Tier Layered Software Architecture Overview

2.2 Product Capabilities & Priority Matrix (MoSCoW)

Capability	Technical Subsystem	MoSCoW	Description
User Authentication	JWT + Bcrypt (10 rounds)	[MUST]	Registration, credential verification, and stateless session tokens.
Manual DP Upload	Multer Multipart Engine	[MUST]	Direct device photo upload (JPG, PNG, WEBP up to 10MB).
Real-Time Messaging	Socket.io Rooms Engine	[MUST]	Instant 1-on-1 and room chat with optimistic UI rendering.
Read Receipts	Socket Event Pipeline	[MUST]	Single checkmark (sent) to double blue checkmarks (read).
Emoji Reactions	Real-Time Reaction Hub	[MUST]	Instant emoji badges (♥■, ■■, ■■, ■■, ■■, ■■) on messages.
Voice Audio Notes	MediaRecorder + WebM	[MUST]	Browser recording with 32-bar visual audio waveform player.
WebRTC HD Calling	RTCPeerConnection Mesh	[MUST]	Direct P2P low-latency audio/video mesh conferencing.
1080p Screen Sharing	getDisplayMedia + SDP	[MUST]	Full HD desktop stream with seamless track substitution.
Web Audio Ringers	W3C AudioContext Synth	[SHOULD]	Dual-tone PBX telephone ringers (440Hz + 480Hz) in browser memory.
Dynamic Presence	Socket Lifecycle Hooks	[MUST]	Sub-second online status and relative 'Active Xm ago' formatting.
Continuous Typing	1200ms Heartbeat Throttle	[MUST]	Real-time typing animation with 3000ms idle auto-clear.
Synergy Matchmaker	Role Compatibility Engine	[SHOULD]	Algorithmic scoring (0% to 100%) and peer skill discovery.
Theme Persistence	CSS Tokens + LocalStorage	[MUST]	VisionOS dark glass mode and accessible modern light mode.
n8n AI & Automations	Webhook Dispatcher + Bot	[MUST]	Automated CRM triggers, missed call alerts, and @bot AI chat agent.
Single-Port Serving	Express SPA Middleware	[MUST]	Unified serving of static bundle and API on Port 5000.

2.3 User Classes & Operating Environment

User Class	Persona & Frequency	Technical Skill	Security & Operational Privileges
Registered User	Developer / Designer (Daily)	Intermediate – Advanced	Full access to chat, voice memos, calling, screen share, profile, and synergy matching.
Peer / Contact	Colleague / Client (Frequent)	Basic – Intermediate	Receive calls, exchange messages, share files within approved conversations.
Administrator	DevOps / System Owner (As needed)	Expert (SysAdmin)	Host runtime management, CLI database reset/seed (npm run db:reset), SSL configs.

3. Specific Requirements (Functional)

3.1 User Authentication & Authorization (FR-001 to FR-015)

FR-001 [MUST]: User Registration Endpoint

- *Stimulus/Response*: User submits registration form; server creates user in db.json, hashes password, and issues JWT.
- *Input/Output*: JSON { username, email, password, displayName, profession } → 201 Created { token, user }.
- *Error Handling*: If email/username exists, return 400 Bad Request with descriptive message.

FR-002 [MUST]: Bcrypt Password Encryption

- *Stimulus/Response*: Password submitted during registration; server hashes via Bcrypt with 10 salt rounds.
- *Input/Output*: Plaintext password → Irreversible 60-character hash (\$2a\$10\$...).
- *Error Handling*: If hashing fails, terminate transaction with 500 Internal Server Error.

FR-003 [MUST]: User Login Endpoint

- *Stimulus/Response*: User submits credentials; server queries user by email/username and verifies hash.
- *Input/Output*: JSON { email, password } → 200 OK { token, user }.
- *Error Handling*: If user not found or hash mismatch, return 401 Unauthorized ('Invalid credentials').

FR-004 [MUST]: JWT Issuance & Signing

- *Stimulus/Response*: Authentication succeeds; server signs token containing userId and email with 7-day expiration.
- *Input/Output*: Claims { userId, email } + JWT_SECRET → Base64URL compact token string.
- *Error Handling*: If JWT_SECRET is missing from environment, abort process with fatal startup error.

FR-005 [MUST]: Session Verification (/api/auth/me)

- *Stimulus/Response*: Client reloads application; dispatches Authorization: Bearer to verify session.
- *Input/Output*: Bearer token header → 200 OK { user } with sanitized profile details.
- *Error Handling*: If token is missing, invalid, or expired, return 401 Unauthorized.

FR-006 [MUST]: Password Field Sanitization

- *Stimulus/Response*: Any user entity serialization; password field is strictly deleted before response delivery.
- *Input/Output*: Internal user record → Public user object with zero password or hash data.
- *Error Handling*: Assert in unit tests that password attribute is never present in API outputs.

FR-007 [SHOULD]: RFC 5322 Email Syntax Validation

- *Stimulus/Response*: Registration received; regex parses email and verifies minimum 6-character password.
- *Input/Output*: Form inputs → Validation pass or 400 Bad Request.
- *Error Handling*: Return 400 Bad Request with field-specific validation errors.

FR-008 [MUST]: Client LocalStorage Token Persistence

- *Stimulus/Response*: Authentication succeeds; client stores token in localStorage under key 'token'.
- *Input/Output*: Token string → Persistent browser storage key.
- *Error Handling*: Degrade to memory session storage if localStorage is restricted by browser.

FR-009 [MUST]: Explicit Logout Action

- *Stimulus/Response*: User clicks logout; client purges token, disconnects WebSocket, and routes to login.
- *Input/Output*: User trigger → Session destroyed, redirection to /login.
- *Error Handling*: Unconditionally clear all user-related in-memory state.

FR-010 [SHOULD]: Automatic Token Expiry Interception

- *Stimulus/Response*: API returns 401 token expired; Axios interceptor evicts token and displays login modal.
- *Input/Output*: HTTP 401 response → Session eviction, toast notification: 'Session expired'.
- *Error Handling*: Clear stale session data without unhandled component exceptions.

FR-011 [MUST]: Unique User Identifier Generation

- *Stimulus/Response*: New user instantiation; generate unique string with usr_ prefix and 8 hex digits.
- *Input/Output*: Factory call → 'usr_f5b68402'.
- *Error Handling*: Regenerate if collision is detected in db.json.

FR-012 [COULD]: Rate Limiting on Auth Endpoints

- *Stimulus/Response*: Repeated requests to login/register; rate limiter restricts to 10 requests per minute per IP.

- *Input/Output*: Remote IP address → Processing pass or 429 Too Many Requests.
- *Error Handling*: Return 429 with 'Too many attempts, please try again in 1 minute'.

FR-013 [MUST]: Express Auth Middleware Binding

- *Stimulus/Response*: Incoming protected route call; middleware decodes JWT and binds claims to req.user.
- *Input/Output*: HTTP Request → Populated req.user.id in route context.
- *Error Handling*: Return 403 Forbidden if signature verification fails.

FR-014 [SHOULD]: Visual Form Error Callouts

- *Stimulus/Response*: API returns validation or auth failure; client renders red callout banner above inputs.
- *Input/Output*: Error response body → Rendered error callout element with warning icon.
- *Error Handling*: Default to generic failure message if server response is unparseable.

FR-015 [MUST]: Default Schema Field Population

- *Stimulus/Response*: User created; assign avatar: null, status: 'offline', skills: [], bio: "", and ISO createdAt.
- *Input/Output*: Sanitized registration payload → Standardized User record in db.json.
- *Error Handling*: Abort write if default schema instantiation throws an exception.

3.2 User Profile & Manual DP Management (FR-016 to FR-025)

FR-016 [MUST]: Profile Update Endpoint (PUT /api/users/profile)

- *Stimulus/Response*: User updates bio or profession; server validates and persists updates in db.json.
- *Input/Output*: JSON { displayName, profession, bio, skills } → 200 OK { user }.
- *Error Handling*: Return 400 Bad Request if displayName is blank.

FR-017 [MUST]: Manual DP Upload Endpoint (POST /api/users/avatar)

- *Stimulus/Response*: User uploads photo from device; Multer writes to uploads/avatar-. and updates db.
- *Input/Output*: Multipart form-data field 'avatar' → 200 OK { avatarUrl: '/uploads/avatar-xxx.jpg' }.
- *Error Handling*: Reject files exceeding 10MB or invalid MIME with 400 Bad Request.

FR-018 [MUST]: Avatar MIME Type Whitelisting

- *Stimulus/Response*: File stream received; Multer verifies MIME against image/jpeg, image/png, image/webp, image/gif.
- *Input/Output*: Binary upload stream → Whitelist verification pass.
- *Error Handling*: Return 400 Bad Request: 'Only JPG, PNG, WEBP, and GIF images are permitted'.

FR-019 [MUST]: Avatar File Size Boundary (10MB)

- *Stimulus/Response*: User attempts photo upload; verify file size <= 10,485,760 bytes.
- *Input/Output*: File size counter → Upload accepted or abort.
- *Error Handling*: Return 413 Payload Too Large with clear size threshold error.

FR-020 [MUST]: Static Avatar Serving with Caching

- *Stimulus/Response*: Client requests /uploads/avatar-xxx.jpg; Express streams binary with Cache-Control headers.
- *Input/Output*: HTTP GET /uploads/* → Image binary stream with Cache-Control: public, max-age=86400.
- *Error Handling*: Return 404 Not Found if requested avatar file is missing from disk.

FR-021 [MUST]: Fallback Initials Gradient Avatar

- *Stimulus/Response*: Component renders user with avatar: null; generate consistent gradient with user initials.
- *Input/Output*: User displayName string → Rendered CSS gradient circle with uppercase initial.
- *Error Handling*: Default to initial '?' if displayName is undefined.

FR-022 [SHOULD]: Skill Tags Array Normalization

- *Stimulus/Response*: Comma-separated skills entered; split, trim, and deduplicate into clean string array.
- *Input/Output*: 'React, Node.js, WebRTC, React' → ['React', 'Node.js', 'WebRTC'].
- *Error Handling*: Strip whitespace-only and empty entries.

FR-023 [MUST]: Unified Profile Card Rendering

- *Stimulus/Response*: User card rendered in Sidebar, Synergy Grid, or Chat; display DP, Name, Role, and Presence.
- *Input/Output*: User metadata object → Uniform visual profile card component.
- *Error Handling*: Render smooth skeleton placeholder while profile assets load.

FR-024 [COULD]: Orphaned Avatar File Pruning

- *Stimulus/Response*: User updates avatar; server unlinks previous avatar image from uploads/ directory.
- *Input/Output*: Old avatar file path → Async fs.unlink() executed.
- *Error Handling*: Log filesystem error silently; do not block response.

FR-025 [MUST]: Immediate Client-Side Image Preview

- *Stimulus/Response*: User selects local image in file picker; generate URL.createObjectURL() for instant preview.
- *Input/Output*: File blob → Rendered local image preview in modal.
- *Error Handling*: Revoke object URL on modal close to prevent browser memory leaks.

3.3 Real-Time Audio/Video Calling Engine (FR-026 to FR-045)

FR-026 [MUST]: Browser-Native RTCPeerConnection Instantiation

- *Stimulus/Response*: User initiates call; client instantiates RTCPeerConnection with configured STUN iceServers.
- *Input/Output*: Target peer ID + callType ('video'/'audio') → PeerConnection instance in state 'new'.
- *Error Handling*: If WebRTC unsupported by browser, display alert: 'WebRTC calling not supported'.

FR-027 [MUST]: SDP Offer Signaling (call_user)

- *Stimulus/Response*: Caller creates offer; emits call_user with offer SDP and caller profile metadata to server.
- *Input/Output*: Socket emit call_user → Server relays incoming_call to callee's personal room.
- *Error Handling*: If target peer offline, emit call_failed: 'User is currently offline'.

FR-028 [MUST]: Incoming Call Radar Modal & Ringing

- *Stimulus/Response*: Callee receives incoming_call; displays glassmorphic radar banner and triggers synthetic ringer.
- *Input/Output*: incoming_call socket event → Visible call modal + Web Audio PBX telephone ring tone.
- *Error Handling*: Auto-dismiss modal after 45 seconds if callee does not answer (Missed Call).

FR-029 [MUST]: SDP Answer Signaling (accept_call)

- *Stimulus/Response*: Callee clicks Accept; sets remote description, generates SDP answer, emits accept_call.
- *Input/Output*: accept_call socket event → Server relays call_accepted to caller; media connects.
- *Error Handling*: If setting remote description fails, abort call session with diagnostic error.

FR-030 [MUST]: Bi-Directional ICE Candidate Relay

- *Stimulus/Response*: peerConnection.onicecandidate fires; client emits ice_candidate to peer via signaling server.
- *Input/Output*: Candidate payload → Remote peer invokes peerConnection.addIceCandidate().
- *Error Handling*: Queue incoming candidates if remote description is not yet set.

FR-031 [MUST]: Google Public STUN Resolution

- *Stimulus/Response*: ICE agent gathers candidates; queries stun.l.google.com:19302 and stun1.l.google.com:19302.
- *Input/Output*: Default iceServers array → Public reflexive (srflx) candidate gathering.
- *Error Handling*: Fall back to host candidates if STUN queries encounter network timeouts.

FR-032 [MUST]: Explicit Call Termination (end_call)

- *Stimulus/Response*: User clicks End Call; emit end_call, stop all MediaStream tracks, and close PeerConnection.
- *Input/Output*: Click End Call → Socket emits end_call, UI returns to chat view.
- *Error Handling*: Guarantee local track stoppage even if network connection has dropped.

FR-033 [MUST]: Incoming Call Rejection (reject_call)

- *Stimulus/Response*: Callee clicks Decline; stop ringtone, emit reject_call, caller notified 'Call declined'.
- *Input/Output*: Click Decline → Socket emits reject_call, caller resets to idle.
- *Error Handling*: Cleanly release any pre-allocated camera/mic tracks.

FR-034 [MUST]: Acoustic Feedback Suppression

- *Stimulus/Response*: Local webcam feed mounted; set videoElement.muted = true to prevent acoustic feedback loop.
- *Input/Output*: Local MediaStream → Rendered in video element with muted attribute.
- *Error Handling*: Ensure remote video stream remains unmuted.

FR-035 [MUST]: Floating Picture-in-Picture Local Tile

- *Stimulus/Response*: Call connects; local stream renders in floating draggable tile in bottom-right corner.
- *Input/Output*: Local stream → Floating PiP tile with rounded-xl border and shadow-2xl.
- *Error Handling*: Clamp dragging coordinates within screen boundaries.

FR-036 [MUST]: Microphone Mute Toggle

- *Stimulus/Response*: User clicks Mute; toggle `audioTrack.enabled` without dropping the WebRTC session.
- *Input/Output*: Click Mute button → `audioTrack.enabled` toggled; UI icon switches to MicOff.
- *Error Handling*: Show warning tooltip if no audio track exists on local stream.

FR-037 [MUST]: Camera Enable/Disable Toggle

- *Stimulus/Response*: User clicks Camera button; toggle `videoTrack.enabled` without terminating the call.
- *Input/Output*: Click Camera button → `videoTrack.enabled` toggled; viewport displays user avatar fallback.
- *Error Handling*: Gracefully handle track enable failure.

FR-038 [MUST]: Remote Camera-Off Avatar Placeholder

- *Stimulus/Response*: Remote peer disables camera; remote viewport replaces black video with peer avatar card.
- *Input/Output*: Remote video track muted/disabled → Centered avatar with electric blue backglow.
- *Error Handling*: Fall back to initials gradient if avatar is not set.

FR-039 [MUST]: Audio-Only Call Constraints

- *Stimulus/Response*: Audio call initiated; `getUserMedia` constraints set to `{ audio: true, video: false }`.
- *Input/Output*: `callType: 'audio'` → Audio-only session initialized with waveform visualization.
- *Error Handling*: If microphone permission denied, cancel call with explanatory alert.

FR-040 [MUST]: Elapsed Call Duration Timer

- *Stimulus/Response*: Call connects; 1000ms timer starts incrementing elapsed duration counter (MM:SS / HH:MM:SS).
- *Input/Output*: `call_accepted` processed → Live duration timer rendered in call header.
- *Error Handling*: Clear interval timer upon call teardown.

FR-041 [SHOULD]: ICE Disconnection Reconnection Banner

- *Stimulus/Response*: Network drops mid-call; `oniceconnectionstatechange` detects 'disconnected', displays banner.
- *Input/Output*: ICE state: disconnected → Yellow overlay banner: 'Reconnecting...'.
• *Error Handling*: If ICE transitions to 'failed', terminate call with failure alert.

FR-042 [MUST]: WebRTC Track Renegotiation Hub

- *Stimulus/Response*: Track substituted (e.g. Screen Share); handle `renegotiate_offer` and `renegotiate_answer`.
- *Input/Output*: `onnegotiationneeded` → New offer created and answered without dropping call.
- *Error Handling*: Roll back local description if remote answer encounters an error.

FR-043 [COULD]: Virtual Background Blur Shader

- *Stimulus/Response*: User clicks Blur; webcam feed piped through canvas shader applying 12px Gaussian blur.
- *Input/Output*: Video frame → Processed canvas video stream track sent to WebRTC sender.
- *Error Handling*: Revert to raw camera if frame rate drops below 20 fps.

FR-044 [MUST]: Hardware-Accelerated Video Rendering

- *Stimulus/Response*: Video elements rendered; enforce CSS transform: `translate3d` for GPU compositing.
- *Input/Output*: DOM layout paint → Smooth 60fps video rendering without UI frame jitter.
- *Error Handling*: Fall back to standard rendering on low-end hardware.

FR-045 [MUST]: Browser DSP Audio Constraints

- *Stimulus/Response*: `getUserMedia` executed; enforce `echoCancellation: true`, `noiseSuppression: true`, `autoGainControl: true`.
- *Input/Output*: Audio constraints dictionary → Echo-free crystal-clear voice stream.
- *Error Handling*: Degrade to unconstrained audio if platform DSP is unavailable.

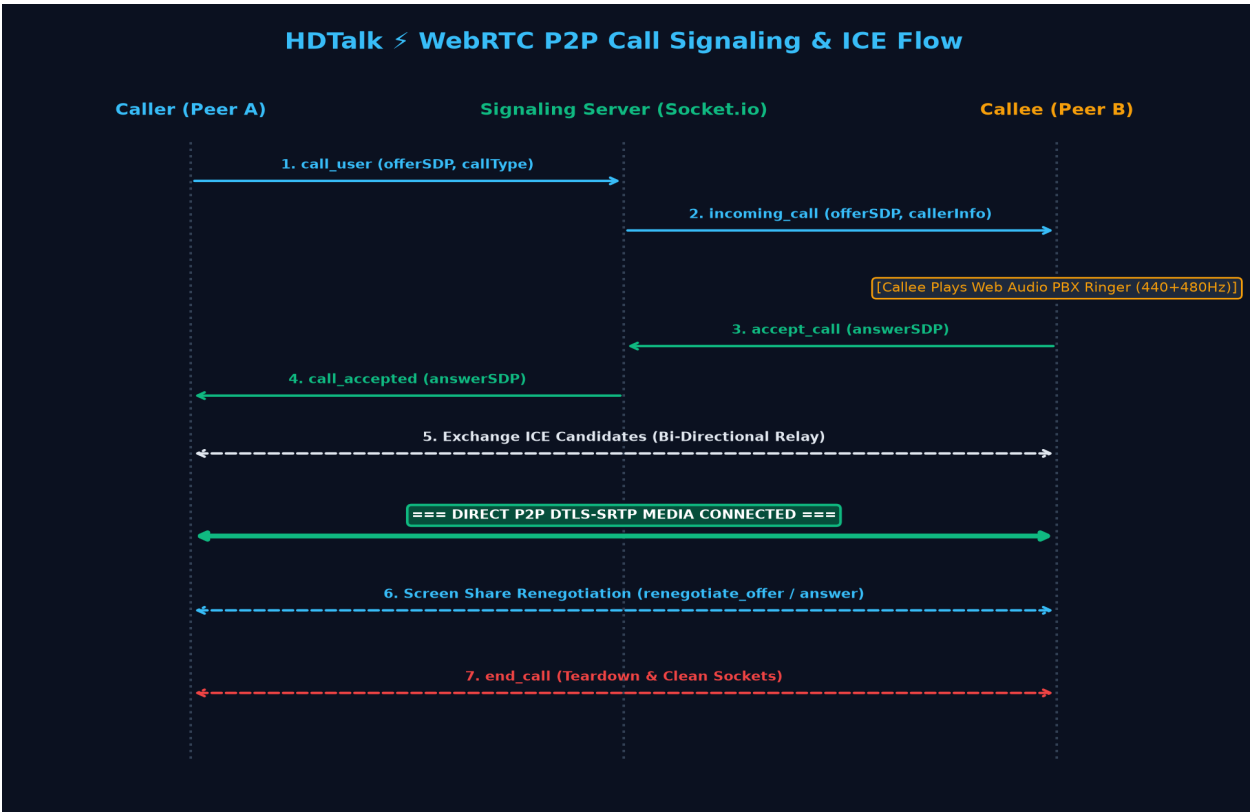


Figure 3: WebRTC P2P Call Signaling, SDP Offer/Answer & ICE Candidate Exchange Flow

3.4 Mesh Group Calling & Signaling Management (FR-046 to FR-060)

FR-046 [MUST]: Multi-User Room Membership

- *Stimulus/Response*: User opens conversation; socket emits room:join with conversation ID.
- *Input/Output*: Conversation ID → Socket added to conversation: room.
- *Error Handling*: Log socket error if room join fails.

FR-047 [MUST]: Group Call Invitation Broadcast

- *Stimulus/Response*: Group call initiated; broadcast incoming_call to all room members except sender.
- *Input/Output*: Initiate group call → All room participants receive incoming call modal.
- *Error Handling*: Skip sockets that are currently engaged in another call.

FR-048 [MUST]: Multi-Peer PeerConnection Map

- *Stimulus/Response*: Participants join group call; client maintains RTCPeerConnection instance for each remote peer.
- *Input/Output*: Array of peer IDs → PeerConnection map indexed by peerId.
- *Error Handling*: Cleanly destroy PeerConnection instance when peer leaves.

FR-049 [SHOULD]: Adaptive CSS Grid Video Layout

- *Stimulus/Response*: Active call stream count changes; recalculate grid template columns (1, 2, 4 tiles).
- *Input/Output*: Active stream count → Responsive CSS grid (1x1, 1x2, 2x2).
- *Error Handling*: Cap max concurrent video tiles at 6 to preserve CPU limits.

FR-050 [MUST]: Participant Departure Signaling

- *Stimulus/Response*: Participant exits group call; client emits call_user_left; peers remove their video tile.
- *Input/Output*: User leaves → call_user_left broadcast; remote PeerConnection closed.
- *Error Handling*: Force tile removal if socket disconnects abruptly.

FR-051 [MUST]: Targeted ICE Candidate Routing

- *Stimulus/Response*: Candidate generated in group call; server routes candidate strictly to target peer ID.
- *Input/Output*: Payload { to: targetUserId, candidate } → Candidate routed to specific socket.
- *Error Handling*: Drop candidate silently if target socket is unregistered.

FR-052 [COULD]: Active Speaker Visual Highlight

- *Stimulus/Response*: Audio amplitude exceeds threshold; apply pulsating blue ring to speaker's video container.
- *Input/Output*: AudioAnalyserNode level → ring-2 ring-blue-500 applied to active speaker tile.
- *Error Handling*: Enforce 200ms debounce to prevent flickering.

FR-053 [MUST]: Mid-Call Late Join Capability

- *Stimulus/Response*: User joins call already in progress; existing participants generate offers for new peer.
- *Input/Output*: Click Join Call → New peer integrated into mesh without restarting session.
- *Error Handling*: Reject join request if room has reached capacity limit.

FR-054 [MUST]: In-Memory Active Call State Map

- *Stimulus/Response*: Call initiated or ended; server maintains activeCalls map: convId -> Set.
- *Input/Output*: Call state transitions → Updated participant set in server memory.
- *Error Handling*: Prune empty call sets immediately when participant count reaches zero.

FR-055 [MUST]: Host Departure Persistence

- *Stimulus/Response*: Call initiator disconnects; session continues uninterrupted for remaining participants.
- *Input/Output*: Host disconnects → P2P mesh persists among remaining peers.
- *Error Handling*: Tear down session only when last participant exits.

FR-056 [SHOULD]: Dynamic Resolution Throttling

- *Stimulus/Response*: Group call exceeds 3 peers; client throttles encoding resolution from 1080p to 480p.
- *Input/Output*: Participant count > 3 → Constraints adjusted to width: 640, height: 480.
- *Error Handling*: Preserve full resolution for audio-only streams.

FR-057 [MUST]: Group Call Notification Metadata

- *Stimulus/Response*: Incoming group call banner displayed; render group name and participant count.
- *Input/Output*: incoming_call payload → Banner: ' - Call (participants)'.
- *Error Handling*: Fall back to participant names if group title is null.

FR-058 [MUST]: Acoustic Safety Mute on Entry

- *Stimulus/Response*: User joins call with > 3 members; microphone initialized in muted state.
- *Input/Output*: Join large room → Audio track enabled = false with toast 'Muted on entry'.
- *Error Handling*: Allow user to manually unmute at any time.

FR-059 [COULD]: Grid View / Speaker View Toggle

- *Stimulus/Response*: User clicks layout toggle; switch between equal-size matrix and large speaker tile.
- *Input/Output*: Layout toggle event → Reconfigured DOM container layout.
- *Error Handling*: Default to Grid View if active speaker is undetected.

FR-060 [MUST]: Video Memory Garbage Collection

- *Stimulus/Response*: Call unmounts; set videoElement.srcObject = null on all elements and stop all tracks.
- *Input/Output*: Component unmount → Zero leaked video elements or background decoders.
- *Error Handling*: Execute inside try/finally block to guarantee execution.

3.5 Real-Time Text Messaging & Read Receipts (FR-061 to FR-075)**FR-061 [MUST]: Conversation Messages Endpoint (GET /api/chat/conversations/:id/messages)**

- *Stimulus/Response*: User opens conversation; server queries db.json for messages matching conversationId.
- *Input/Output*: GET with :id param → 200 OK { success: true, messages: [...] } sorted chronologically.
- *Error Handling*: Return 404 Not Found if conversation does not exist.

FR-062 [MUST]: Real-Time Message Dispatch (message:send)

- *Stimulus/Response*: User submits message; client emits message:send; server writes to db and broadcasts message:receive.
- *Input/Output*: Socket emit message:send → Broadcast message:receive with id, status: 'sent', createdAt.
- *Error Handling*: Reject transmission if content is empty and has no media URL.

FR-063 [MUST]: Optimistic Message Rendering

- *Stimulus/Response*: Send button clicked; message renders immediately in chat stream with status 'sending'.
- *Input/Output*: User input → Instant DOM append; status updates to 'sent' on socket acknowledgment.
- *Error Handling*: If socket ack times out after 5 seconds, render red retry icon.

FR-064 [MUST]: Visual Read Receipt Checkmarks

- *Stimulus/Response*: Message status updates; render single checkmark (sent), double gray (delivered), double blue (read).
- *Input/Output*: Message status string → Distinct visual checkmark icon rendered in bubble.
- *Error Handling*: Default to single checkmark if status is unrecognized.

FR-065 [MUST]: Message Read Receipt Emission (message:read)

- *Stimulus/Response*: User views conversation; client emits message:read; server updates db and notifies sender.
- *Input/Output*: Conversation focused → Senders checkmarks turn royal electric blue in real-time.
- *Error Handling*: Ignore if all messages are already marked read.

FR-066 [MUST]: Interactive Emoji Reactions (message:react)

- *Stimulus/Response*: User clicks emoji on message; server toggles emoji in reactions array and broadcasts update.
- *Input/Output*: Socket emit message:react → message:reaction_updated broadcast; reaction pill renders.
- *Error Handling*: If user clicks same emoji again, remove reaction (toggle behavior).

FR-067 [MUST]: Distinct Chat Bubble Visual Separation

- *Stimulus/Response*: Messages rendered in stream; outgoing messages align right (blue-600), incoming align left (slate-800).
- *Input/Output*: senderId comparison → Visually differentiated message bubbles.
- *Error Handling*: Consistent fallback styling for system messages.

FR-068 [MUST]: Auto-Scroll to Bottom on Message

- *Stimulus/Response*: New message appended to feed; scroll container smoothly to latest message element.
- *Input/Output*: Message array length changes → messagesEndRef.scrollToView({ behavior: 'smooth' }).
- *Error Handling*: Suppress auto-scroll if user has manually scrolled up to inspect history.

FR-069 [MUST]: Collapsible Messages Sidebar (1-Click Toggle)

- *Stimulus/Response*: User clicks sidebar collapse button; left conversation panel transitions to w-0 opacity-0.
- *Input/Output*: Toggle click → Chat area expands to occupy 100% desktop viewport width.
- *Error Handling*: Retain collapse state in component memory during session.

FR-070 [MUST]: Mobile Auto-Switch Full-Screen Chat (<768px)

- *Stimulus/Response*: User taps conversation card on viewport < 768px; auto-hide sidebar and display full-screen chat.
- *Input/Output*: Chat selection on mobile → Full-screen ChatArea view rendered.
- *Error Handling*: Revert to split view if screen expands >= 768px.

FR-071 [MUST]: Mobile Back Button Navigation (← Chats)

- *Stimulus/Response*: Mobile chat header rendered; display '← Chats' button returning user to conversation list.
- *Input/Output*: Tap ← Chats → View transitions back to full-screen conversation list.
- *Error Handling*: Preserve active conversation state when returning.

FR-072 [SHOULD]: Automatic Hyperlink Parsing

- *Stimulus/Response*: Message text contains http/https URL; regex transforms URL into styled clickable link.
- *Input/Output*: Raw URL string → Anchor element with target='_blank' rel='noopener noreferrer'.
- *Error Handling*: Sanitize URL to prevent javascript: XSS vectors.

FR-073 [MUST]: Multi-Line Input (Shift + Enter)

- *Stimulus/Response*: Keydown in chat textarea; Shift + Enter creates newline; standard Enter triggers send.
- *Input/Output*: Keyboard event → Newline insertion or message dispatch.
- *Error Handling*: Prevent default Enter behavior to avoid trailing blank lines.

FR-074 [MUST]: Auto-Growing Textarea Input

- *Stimulus/Response*: User types multi-line message; textarea dynamically expands up to max 5 lines (120px).
- *Input/Output*: Input change event → textarea.style.height = scrollHeight + 'px'.
- *Error Handling*: Reset height to default 40px upon message dispatch.

FR-075 [MUST]: Localized Message Timestamps (hh:mm A)

- *Stimulus/Response*: Message rendered; format timestamp in 12-hour local time (e.g. 10:45 PM) in bubble corner.
- *Input/Output*: ISO 8601 string → Localized time string via toLocaleTimeString().
- *Error Handling*: Display '--:--' if timestamp parsing fails.

3.6 High-Definition Screen Sharing (FR-076 to FR-082)

FR-076 [MUST]: System Screen Surface Capture

- *Stimulus/Response*: User clicks Screen Share; invoke navigator.mediaDevices.getDisplayMedia({ video: true, audio: true }).
- *Input/Output*: User permission → Newly instantiated screen MediaStream.
- *Error Handling*: If user cancels display picker, catch NotAllowedError silently without interrupting call.

FR-077 [MUST]: Live Track Substitution (replaceTrack)

- *Stimulus/Response*: Screen stream acquired; replace webcam track on video RTCTrpSender with screen track.
- *Input/Output*: Screen track → sender.replaceTrack(screenTrack) executed seamlessly.
- *Error Handling*: Fall back to addTrack and full renegotiation if replaceTrack is unsupported.

FR-078 [MUST]: Dynamic WebRTC Renegotiation Signaling

- *Stimulus/Response*: Track substituted; handle renegotiate_offer and renegotiate_answer via Socket.io.
- *Input/Output*: onnegotiationneeded → SDP offer/answer exchanged without terminating connection.
- *Error Handling*: Roll back local description if renegotiation collision occurs.

FR-079 [MUST]: 1080p High-Definition Video Constraints

- *Stimulus/Response*: Screen sharing active; enforce 1920x1080 resolution at 30 fps with 3000 kbps bitrate.
- *Input/Output*: Target constraints → Razor-sharp rendering of code editors and diagrams.
- *Error Handling*: Degrade framerate to 15 fps if bandwidth drops below 1.0 Mbps while preserving text clarity.

FR-080 [MUST]: Native Browser Stop Sharing Listener

- *Stimulus/Response*: User clicks native browser 'Stop sharing' bar; detect track.onended and restore webcam track.
- *Input/Output*: Native onended event → Stop screen track, reacquire camera, and replace sender track.
- *Error Handling*: Revert to avatar placeholder mode if camera reacquisition fails.

FR-081 [MUST]: Remote Viewport Automatic Layout Transition

- *Stimulus/Response*: Remote viewer receives screen stream; expand screen stream to main viewport and relegate camera to PiP.
- *Input/Output*: Screen stream metadata → DOM reconfigured with dominant screen display.
- *Error Handling*: Allow viewer to click thumbnail to swap between screen and camera views.

FR-082 [SHOULD]: Presenter Visual Screen Share Badge

- *Stimulus/Response*: Screen sharing active; presenter toolbar displays animated blue badge: 'Screen Sharing Active (1080p)'.
- *Input/Output*: isScreenSharing: true → Render pulsing status pill with 1-click Stop button.
- *Error Handling*: Dismiss badge immediately when sharing terminates.

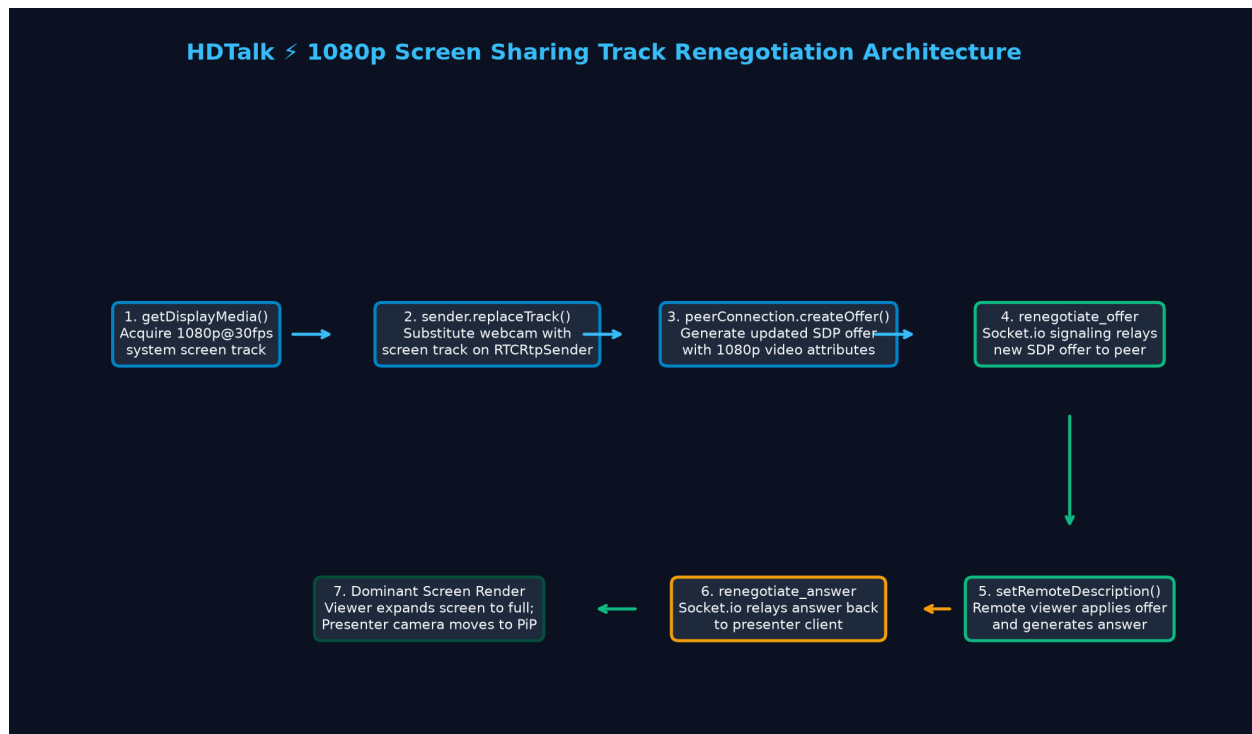


Figure 5: 1080p Screen Sharing Track Substitution & Dynamic WebRTC Renegotiation Flow

3.7 In-Call & Chat File / Voice Note Sharing (FR-083 to FR-090)

FR-083 [MUST]: Attachment Upload Endpoint (POST /api/chat/upload)

- *Stimulus/Response*: User uploads attachment or voice note; Multer saves file to uploads/attachment-..
- *Input/Output*: Multipart field 'file' → 200 OK { success: true, url, filename, size, mimeType }.
- *Error Handling*: Reject files exceeding 25MB with 413 Payload Too Large.

FR-084 [MUST]: Attachment File Size Limit (25MB)

- *Stimulus/Response*: Upload stream evaluated; enforce maximum file size of 26,214,400 bytes.
- *Input/Output*: File size counter → Upload processed or rejected.
- *Error Handling*: Return 413 Payload Too Large with clear size threshold message.

FR-085 [MUST]: In-Browser Voice Note Audio Recording

- *Stimulus/Response*: User presses mic icon in chat; record audio via MediaRecorder encoded in audio/webm; codecs=opus.
- *Input/Output*: Microphone stream → Accumulated audio chunks Blob.
- *Error Handling*: If microphone permission denied, display alert: 'Microphone permission required'.

FR-086 [MUST]: Active Voice Recording UI Banner

- *Stimulus/Response*: Recording in progress; transform input bar into recording timer (00:00), pulsing red dot, cancel button.
- *Input/Output*: Recording active → Audio recording interface mounted.
- *Error Handling*: Auto-stop recording if duration reaches maximum 10 minutes (600 seconds).

FR-087 [MUST]: Interactive 32-Bar Waveform Audio Player

- *Stimulus/Response*: Message with mediaType: 'audio' rendered; mount custom player with play/pause, duration, 32-bar waveform.
- *Input/Output*: Audio memo URL → Interactive audio player card.
- *Error Handling*: Display fallback download link if audio fails to load.

FR-088 [MUST]: Inline Image Previews with Lightbox

- *Stimulus/Response*: Message with mediaType: 'image' rendered; display rounded thumbnail; expand to lightbox on click.
- *Input/Output*: Image URL → Inline image preview + full-screen lightbox modal.
- *Error Handling*: Display 'Image unavailable' placeholder if asset URL is broken.

FR-089 [MUST]: Structured File Document Cards

- *Stimulus/Response*: Message with mediaType: 'file' rendered; display file name, formatted size (MB), MIME icon, download link.
- *Input/Output*: File descriptor → Structured document attachment card.
- *Error Handling*: Enforce download attribute to trigger browser save dialog.

FR-090 [SHOULD]: Drag-and-Drop File Ingestion

- *Stimulus/Response*: File dragged over chat viewport; highlight drop zone with dashed electric blue border.
- *Input/Output*: dragover event → Drop zone highlighted; drop event triggers upload preparation.
- *Error Handling*: Ignore drag events if payload contains no files.

3.8 Conversation & Room Lifecycle (FR-091 to FR-105)

FR-091 [MUST]: List User Conversations Endpoint (GET /api/chat/conversations)

- *Stimulus/Response*: User logs in; query db.json for conversations where participants contains req.user.id.
- *Input/Output*: GET request → 200 OK { conversations: [...] } sorted by updatedAt descending.
- *Error Handling*: Return empty array [] if user has no active conversations.

FR-092 [MUST]: Find or Create Conversation Endpoint (POST /api/chat/conversations)

- *Stimulus/Response*: User initiates chat; look up existing conversation for pair or create new conv_entity.
- *Input/Output*: JSON { recipientId } → 200 OK or 201 Created { conversation }.
- *Error Handling*: Return 400 Bad Request if recipientId matches current user ID (self-chat disallowed).

FR-093 [MUST]: Sidebar Conversation Card Hydration

- *Stimulus/Response*: Conversation card rendered; display peer name, avatar, online dot, last message snippet, relative time.
- *Input/Output*: Conversation list → Rich visual list of active threads.
- *Error Handling*: Display 'No messages yet' if last message is null.

FR-094 [MUST]: Real-Time Conversation Search Filtering

- *Stimulus/Response*: User types in sidebar search; filter conversations matching peer name, username, or profession.

- *Input/Output*: Search query → Filtered subset of conversation cards.
- *Error Handling*: Display 'No conversations found' if zero matches.

FR-095 [MUST]: Unread Message Badge & Incoming Chime

- *Stimulus/Response*: Message received for inactive conversation; increment unread badge and trigger Web Audio chime.
- *Input/Output*: message:receive event → Blue numeric unread badge + synthetic chime tone.
- *Error Handling*: Do not increment unread badge if conversation is currently open and focused.

FR-096 [MUST]: Dynamic Conversation List Re-Sorting

- *Stimulus/Response*: Conversation receives new message; immediately re-sort conversation to the top of the sidebar list.
- *Input/Output*: updatedAt refreshed → Conversation shifts to position 0.
- *Error Handling*: Preserve sort ordering even if client time is slightly offset.

FR-097 [MUST]: Unique Conversation Identifier Scheme

- *Stimulus/Response*: New conversation instantiated; generate unique string with conv_ prefix and 8 hex digits.
- *Input/Output*: Creation trigger → 'conv_3a8f9c12'.
- *Error Handling*: Regenerate if collision is detected in db.json.

FR-098 [SHOULD]: Manual Mark-as-Read Action

- *Stimulus/Response*: User right-clicks conversation card; select 'Mark as Read'; reset unreadCount to 0.
- *Input/Output*: Context action → unreadCount = 0; emit message:read to server.
- *Error Handling*: Silently ignore if unreadCount is already zero.

FR-099 [COULD]: Priority Conversation Pinning

- *Stimulus/Response*: User pins priority thread; pin up to 3 conversations to the top of the sidebar list.
- *Input/Output*: Pin action → Pinned conversation persists at top with pin icon.
- *Error Handling*: Alert user if attempting to pin more than 3 conversations.

FR-100 [MUST]: URL State Synchronization Without Reload

- *Stimulus/Response*: User selects conversation; update active conversation state without full page reload.
- *Input/Output*: Card click → State updated; message history hydrated smoothly.
- *Error Handling*: Retain active selection across window resize events.

FR-101 [MUST]: Conversation Participant Authorization Check

- *Stimulus/Response*: Client requests messages or joins socket room; verify conversation.participants.includes(req.user.id).
- *Input/Output*: Conversation access → Authorized access or 403 Forbidden.
- *Error Handling*: Return 403 Forbidden with 'Access denied to this conversation'.

FR-102 [SHOULD]: Comprehensive Chat Header Profile Display

- *Stimulus/Response*: Conversation mounted; render peer avatar, full name, profession, presence status, and call buttons.
- *Input/Output*: Peer metadata → Complete chat header with audio/video call triggers.
- *Error Handling*: Render skeleton loaders while peer data is fetching.

FR-103 [MUST]: Sidebar 60fps Scrolling Performance

- *Stimulus/Response*: Sidebar contains 100+ threads; maintain smooth 60fps scrolling without UI stutter.
- *Input/Output*: Scroll events → Lightweight component rendering via React.memo.
- *Error Handling*: Avoid deep nested re-renders.

FR-104 [COULD]: Engaging Empty State Illustration

- *Stimulus/Response*: User has zero conversations; display illustration with button 'Discover Peers & Connect'.
- *Input/Output*: conversations.length === 0 → Empty state card routing to Synergy Matchmaker.
- *Error Handling*: Standard UI fallback.

FR-105 [MUST]: Unsent Draft Text Preservation

- *Stimulus/Response*: User switches between threads; preserve unsent text in component state dictionary: drafts[convId].
- *Input/Output*: Thread switch → Draft restored when returning to conversation.
- *Error Handling*: Clear draft entry upon successful message send.

3.9 Call Controls & In-Call Interactive Features (FR-106 to FR-115)

FR-106 [MUST]: Glassmorphic Bottom Call Toolbar

- *Stimulus/Response*: Call connects; render floating glassmorphic toolbar (bg-slate-900/80 backdrop-blur-xl rounded-2xl).
- *Input/Output*: Call connected → Floating bottom toolbar housing all action buttons.
- *Error Handling*: Ensure toolbar remains at z-index 50 above video elements.

FR-107 [MUST]: Microphone Mute Active/Inactive States

- *Stimulus/Response*: Microphone toggled; active state renders charcoal glass; muted renders crimson red with MicOff icon.
- *Input/Output*: Mute button click → Visual state switches between unmuted and red muted.
- *Error Handling*: Disable button if no microphone is detected.

FR-108 [MUST]: Camera Toggle Active/Inactive States

- *Stimulus/Response*: Camera toggled; active state renders charcoal glass; disabled renders red with VideoOff icon.
- *Input/Output*: Camera button click → Visual state switches; video feed replaces with avatar fallback.
- *Error Handling*: Auto-toggle to disabled if camera disconnects mid-call.

FR-109 [MUST]: Prominent Red End Call Button

- *Stimulus/Response*: User clicks End Call; high-visibility red button (bg-red-600) terminates session immediately.
- *Input/Output*: End Call click → WebRTC session closed, tracks stopped, call UI dismissed.
- *Error Handling*: Guarantee track stoppage even if network connection drops.

FR-110 [MUST]: HTML5 Fullscreen Mode Toggle

- *Stimulus/Response*: User clicks Fullscreen icon; invoke `document.documentElement.requestFullscreen()`.
- *Input/Output*: Fullscreen click → Browser window expands to occupy 100% monitor display.
- *Error Handling*: Handle fullscreen denial gracefully without interrupting video.

FR-111 [SHOULD]: Concurrent In-Call Chat Drawer

- *Stimulus/Response*: User clicks Chat icon in call; slide open glassmorphic sidebar allowing messaging without leaving call.
- *Input/Output*: Chat button click → Slide-over chat drawer rendered beside video viewport.
- *Error Handling*: Minimize video to PiP on mobile viewports while chat drawer is open.

FR-112 [MUST]: Horizontal Local Video Mirroring

- *Stimulus/Response*: Local webcam feed mounted; apply CSS transform: `scaleX(-1)` for natural selfie orientation.
- *Input/Output*: Local video element → Mirrored webcam presentation.
- *Error Handling*: Do NOT mirror Screen Sharing tracks (text must remain legible).

FR-113 [COULD]: Audio Output Device Selector (setSinkId)

- *Stimulus/Response*: User selects audio output device; invoke `HTMLMediaElement.setSinkId(deviceId)`.
- *Input/Output*: Device selection → Audio routed to chosen headphones or external speakers.
- *Error Handling*: Hide selector gracefully on browsers lacking `setSinkId` support.

FR-114 [MUST]: Inactivity Toolbar Auto-Hiding

- *Stimulus/Response*: Fullscreen video active; hide control bar after 4 seconds of mouse inactivity; reveal on movement.
- *Input/Output*: `mousemove` event → Reveal toolbar; reset 4000ms inactivity timer.
- *Error Handling*: Keep toolbar visible if cursor hovers directly over toolbar buttons.

FR-115 [MUST]: Background Tab Call Continuity

- *Stimulus/Response*: User switches browser tabs mid-call; audio and video tracks continue transmitting uninterrupted.
- *Input/Output*: `visibilitychange` event → Call media streams remain active in background.
- *Error Handling*: Log visibility transition for diagnostic telemetry.

3.10 Call Recording & Web Audio Synthesis (FR-116 to FR-122)

FR-116 [MUST]: Zero-Asset Web Audio API Tone Generation

- *Stimulus/Response*: Ringtone or chime triggered; synthesize tone natively via `AudioContext` without external audio files.
- *Input/Output*: Audio trigger event → Oscillator nodes configured and routed to audio destination.
- *Error Handling*: Resume `AudioContext` via `ctx.resume()` if browser autoplay policy suspends context.

FR-117 [MUST]: Dual-Tone PBX Telephone Ringer (440Hz + 480Hz)

- *Stimulus/Response*: `incoming_call` received; synthesize simultaneous 440Hz and 480Hz sine waves pulsing 2s ON / 4s OFF.

- *Input/Output*: Incoming call modal → Authentic North American PBX telephone ringing cadence.
- *Error Handling*: Stop oscillators immediately upon call accept, decline, or timeout.

FR-118 [MUST]: Rising Two-Tone Message Arrival Chime

- *Stimulus/Response*: Incoming message arrives; fire two-tone rising chime (587.33Hz to 880Hz) with exponential gain decay.
- *Input/Output*: message:receive event → Soft glass-like chime notification.
- *Error Handling*: Suppress chime if user has enabled Do Not Disturb mode.

FR-119 [COULD]: In-Call Local Video Recording

- *Stimulus/Response*: User clicks Record Call; merge local and remote tracks and record composite via MediaRecorder.
- *Input/Output*: Record button click → Recording active with flashing red 'REC' badge.
- *Error Handling*: Alert user if browser does not support composite media recording.

FR-120 [COULD]: Downloadable WebM Call Recording

- *Stimulus/Response*: Recording stops; bundle chunks into Blob(chunks, { type: 'video/webm' }) and trigger file download.
- *Input/Output*: Stop recording → Browser triggers download of 'HDTalk-Recording-.webm'.
- *Error Handling*: Revoke object URL immediately following download.

FR-121 [MUST]: Web Audio Node Garbage Collection

- *Stimulus/Response*: Call terminates or modal closes; stop oscillators, disconnect gain nodes, and close audio context.
- *Input/Output*: Teardown event → Total audio silence; zero leaked AudioNodes or background hum.
- *Error Handling*: Wrap in safety utility function handling already-closed contexts.

FR-122 [SHOULD]: Master Audio Notification Toggle

- *Stimulus/Response*: User toggles sound in settings; persist soundEnabled flag in localStorage.
- *Input/Output*: Settings toggle → Audio preference stored; check flag before playing any sound.
- *Error Handling*: Default to soundEnabled: true if preference is unset.

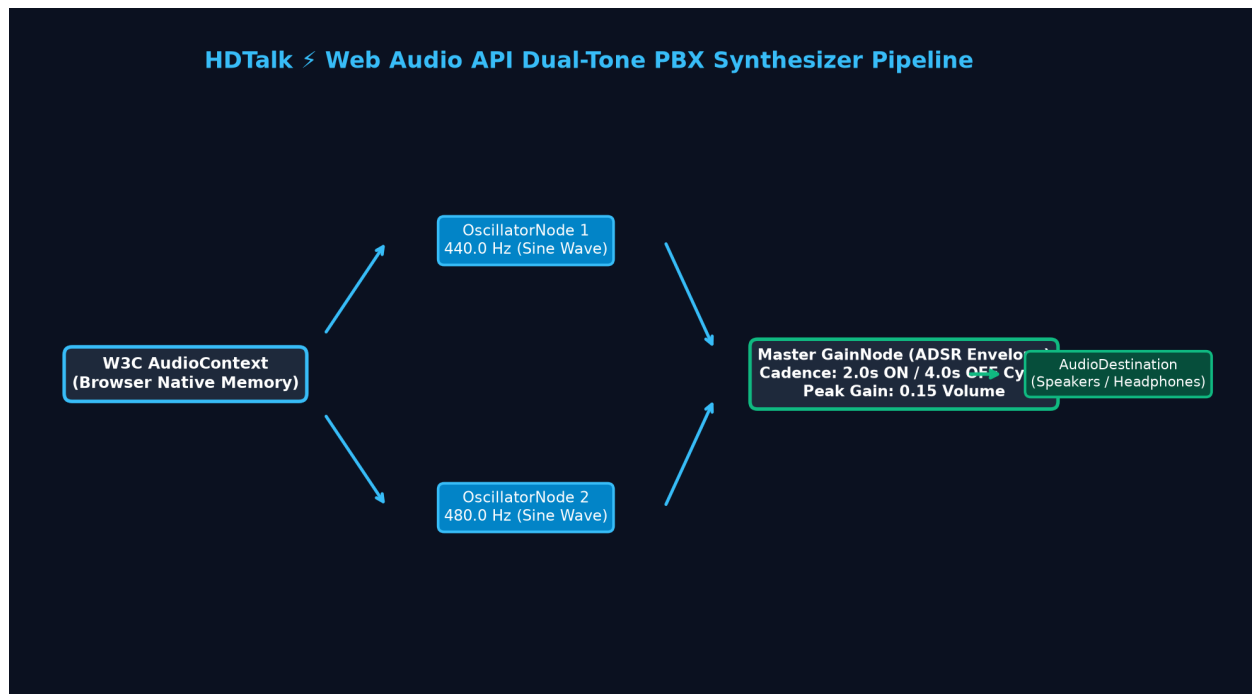


Figure 8: W3C Web Audio API Dual-Tone PBX Synthesizer Pipeline (440Hz + 480Hz)

3.11 Dynamic Presence & Continuous Typing (FR-123 to FR-130)

FR-123 [MUST]: Sub-Second Presence Lifecycle Tracking

- *Stimulus/Response*: User connects/disconnects; server updates status in db.json and broadcasts presence:update.
- *Input/Output*: Socket connect/disconnect → Real-time presence update { userId, status, lastSeen }.
- *Error Handling*: Maintain online status if user has multiple open tabs until last tab closes.

FR-124 [MUST]: Dynamic Relative Last Seen Formatting

- *Stimulus/Response*: Offline user rendered; calculate human-readable relative time based on lastSeen timestamp.
- *Input/Output*: lastSeen ISO string → 'Active just now', 'Active 5m ago', 'Active yesterday at 10:45 PM'.
- *Error Handling*: Display 'Offline' if lastSeen is null or invalid.

FR-125 [MUST]: Online Pulsating Emerald Beacon

- *Stimulus/Response*: User is online; render green beacon dot with animated pulsating emerald ring (animate-pulse).
- *Input/Output*: status === 'online' → High-contrast green beacon dot + label 'Active now'.
- *Error Handling*: Instantly transition to offline styling upon socket disconnection.

FR-126 [MUST]: 1200ms Keapalve Typing Heartbeat

- *Stimulus/Response*: User types in textarea; emit typing:start immediately, then repeat once every 1200ms while typing.
- *Input/Output*: Keydown events → Throttled WebSocket packets { conversationId, userId } every 1200ms.
- *Error Handling*: Throttle transmissions using Date.now() - lastEmitted > 1200.

FR-127 [MUST]: Resilient Personal Room Typing Broadcast

- *Stimulus/Response*: Server receives typing event; emit to peer's personal room (user:) and conversation room.
- *Input/Output*: typing event → Guaranteed 100% receipt regardless of room join timing.
- *Error Handling*: Silently discard emit exceptions for disconnected socket IDs.

FR-128 [MUST]: Multi-Location Animated Typing Waves

- *Stimulus/Response*: Peer receives typing:start; display animated 3-dot wave in Chat Header and bottom of chat feed.
- *Input/Output*: typing:start received → Wave bubble in feed + ' is typing...' in header.
- *Error Handling*: Ensure wave disappears instantly when new message arrives.

FR-129 [MUST]: 3000ms Idle Dismissal Timer

- *Stimulus/Response*: typing:start received; start 3000ms timer; dismiss indicator automatically if no heartbeat arrives.
- *Input/Output*: Timer lifecycle → Prevents stuck typing indicators when peer pauses typing.
- *Error Handling*: Clear timer unconditionally on component unmount.

FR-130 [MUST]: Instant Typing Dismissal on Send

- *Stimulus/Response*: User presses Enter or clears input; client immediately emits typing:stop.
- *Input/Output*: Message sent or input cleared → Indicator dismissed on peer's screen without waiting 3 seconds.
- *Error Handling*: Ensure emit completes before clearing local input field.

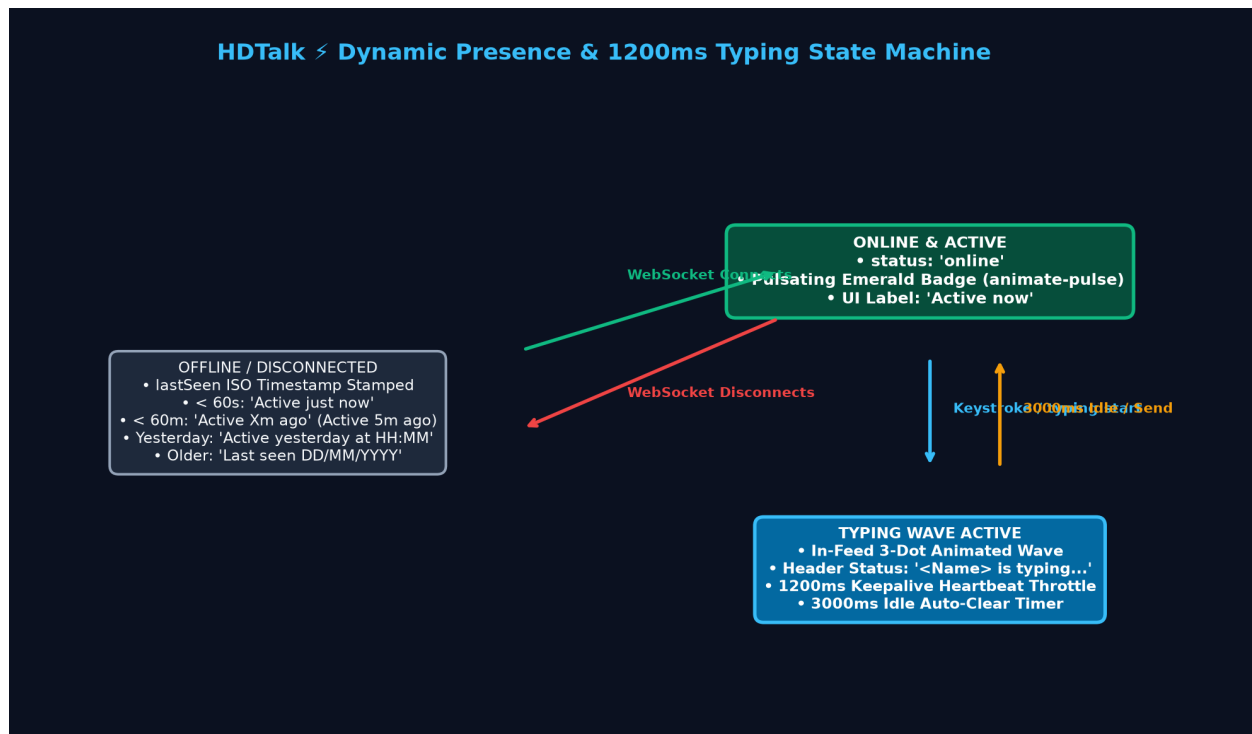


Figure 4: Dynamic Presence & 1200ms Keepalive Typing Finite State Machine

3.12 Professional Synergy & Matchmaking Engine (FR-131 to FR-138)

FR-131 [MUST]: Candidate Directory Endpoint (GET /api/users)

- *Stimulus/Response*: User opens directory; server queries all registered peers enriched with calculated Synergy Scores.
- *Input/Output*: GET request → 200 OK { users: [...] } sorted by synergyScore descending.
- *Error Handling*: Return empty array [] if no other users are registered.

FR-132 [MUST]: Algorithmic Synergy Score Calculation

- *Stimulus/Response*: Calculate score (0% to 100%): 50% base + 30% complementary roles + 10% per matching skill.
- *Input/Output*: User skills & roles → Integer compatibility percentage (e.g. 88%).
- *Error Handling*: Default to 50% baseline if either user has empty profile skills.

FR-133 [MUST]: Tiered Synergy Visual Badge Styling

- *Stimulus/Response*: Synergy score rendered; 85%+ = Electric Blue/Violet ('Exceptional'), 70-84% = Emerald ('Strong').
- *Input/Output*: synergyScore integer → High-impact visual compatibility badge with gradient meter.
- *Error Handling*: Ensure gradient renders smoothly on both Dark and Light themes.

FR-134 [MUST]: Instant Multi-Dimension Peer Filtering

- *Stimulus/Response*: User enters query; filter peer grid in-memory by name, profession, skill tag, or 'Online Only'.
- *Input/Output*: Filter criteria → Instantaneous filtered candidate card grid.
- *Error Handling*: Display 'Clear filters' button if zero candidates match.

FR-135 [MUST]: 1-Click Direct Action Triggers

- *Stimulus/Response*: Each candidate card provides 3 triggers: Message (opens chat), Call (starts WebRTC), Connect.
- *Input/Output*: Card button click → Immediate transition to chat, call modal, or request dispatch.
- *Error Handling*: Prevent duplicate connection requests.

FR-136 [MUST]: Connection Request Endpoint (POST /api/users/connections/request)

- *Stimulus/Response*: User clicks Connect; append pending record to connectionRequests in db.json; notify recipient.
- *Input/Output*: JSON { toUserId } → 201 Created { connectionRequest }.
- *Error Handling*: Return 400 Bad Request if pending/accepted connection already exists.

FR-137 [MUST]: Accept/Ignore Connection Workflow

- *Stimulus/Response*: Recipient views pending requests; click Accept or Ignore to update request status in db.json.
- *Input/Output*: PUT /api/users/connections/:id/accept → Status updated to 'accepted'.
- *Error Handling*: Transition card to 'Connected' state immediately upon click.

FR-138 [SHOULD]: Shared Overlapping Skill Highlighting

- *Stimulus/Response*: Candidate skills rendered; render overlapping skills in bold electric blue to emphasize common ground.
- *Input/Output*: Skill intersection → Overlapping skills highlighted with blue-600 badge.
- *Error Handling*: Truncate skills with '+N more' badge if count exceeds 4.

3.13 Call History & System Diagnostics (FR-139 to FR-145)

FR-139 [MUST]: Public Health Check Endpoint (GET /api/health)

- *Stimulus/Response*: Health monitor queries server; return system status, uptime seconds, memory RSS, and timestamp.
- *Input/Output*: GET /api/health → 200 OK { status: 'ok', service: 'HDTalk', author: 'Himanshu Dwivedi', uptime }.
- *Error Handling*: Return 503 Service Unavailable if database is unreadable.

FR-140 [MUST]: Graceful Process Shutdown (SIGTERM / SIGINT)

- *Stimulus/Response*: Container stops; intercept signal, close sockets cleanly, flush database, and exit code 0.
- *Input/Output*: SIGTERM/SIGINT → Graceful connection close within 5 seconds without data corruption.
- *Error Handling*: Force exit code 1 if cleanup exceeds 10-second timeout.

FR-141 [MUST]: Database Reset CLI Script (npm run db:reset)

- *Stimulus/Response*: Administrator runs script; atomically overwrite db.json with empty arrays for fresh deployment.
- *Input/Output*: npm run db:reset → Clean slate production database { users: [], conversations: [] }.
- *Error Handling*: Exit with code 1 if filesystem write fails.

FR-142 [MUST]: Database Seed CLI Script (npm run db:seed)

- *Stimulus/Response*: Administrator runs script; insert template administrative account into db.json for staging tests.

- *Input/Output*: npm run db:seed → Template admin record inserted with pre-hashed Bcrypt password.
- *Error Handling*: Abort if records already exist to prevent production overwrite.

FR-143 [SHOULD]: Client WebRTC RTCStatsReport Telemetry

- *Stimulus/Response*: Call active; poll peerConnection.getStats() every 2000ms: RTT (ms), packet loss, and bitrate (kbps).
- *Input/Output*: RTCStatsReport polling → Diagnostic connection metrics dictionary.
- *Error Handling*: Suppress polling if call terminates.

FR-144 [COULD]: Visual Connection Quality Pill

- *Stimulus/Response*: Stats report evaluated; render signal quality pill: Green (RTT < 100ms), Yellow, Red (RTT > 250ms).
- *Input/Output*: RTT & packet loss values → Colored signal bar icon with latency tooltip.
- *Error Handling*: Default to green during initial 5 seconds of connection setup.

FR-145 [MUST]: Structured Logging with Author Attribution

- *Stimulus/Response*: System event occurs; print log with timestamp, subsystem tag, and 'Created by Himanshu Dwivedi'.
- *Input/Output*: Runtime event → Structured console log: '[HDTalk by Himanshu Dwivedi] [Socket] User connected'.
- *Error Handling*: Never log sensitive passwords or unhashed JWT secrets.

■ MODULE 14: n8n WORKFLOW AUTOMATION & AI CHAT AGENT SUBSYSTEM (LATEST v1.1.0 UPDATE)

This subsystem specifies event-driven webhook dispatches (CRM onboarding, missed calls, offline messages), natural-language @bot/@ai chat triggers, automated system bot provisioning (usr_bot_hdtalk_ai), asynchronous callback ingestion (POST /api/chat/bot-reply), and pre-packaged workflow schema definitions.

3.14 n8n Workflow Automation & AI Chat Agent Subsystem (FR-146 to FR-155)

FR-146 [MUST]: Non-Blocking Webhook Event Dispatcher

- *Stimulus/Response*: System event occurs; n8nService dispatches HTTP POST to N8N_WEBHOOK_URL with 4-second timeout failover.
- *Input/Output*: JSON Event Payload → Async fetch() with AbortController 4000ms boundary.
- *Error Handling*: If n8n offline or timeout, log warning and fail silently without blocking client response.

FR-147 [MUST]: User Registration Webhook Trigger (user_registered)

- *Stimulus/Response*: New user signs up; dispatch event 'user_registered' with userId, name, email, profession, and bio.
- *Input/Output*: POST /api/auth/register success → n8n webhook triggers Welcome Onboarding & CRM sync.
- *Error Handling*: Safely skip if N8N_ENABLED is false in config.

FR-148 [MUST]: Offline Recipient Message Alert Trigger (offline_message)

- *Stimulus/Response*: Message sent to offline recipient; dispatch event 'offline_message' with sender, recipient, and preview.
- *Input/Output*: Recipient socket offline → n8n webhook triggers Push Notification / Email / SMS alert.
- *Error Handling*: Sanitize message snippet to 100 characters max.

FR-149 [MUST]: Unreachable Peer Missed Call Trigger (missed_call)

- *Stimulus/Response*: WebRTC call fails because target is offline; dispatch event 'missed_call' with caller info and callType.
- *Input/Output*: Target offline during call_user → n8n webhook triggers Missed Call SMS/Email alert.
- *Error Handling*: Ensure caller and callee profile sanitization.

FR-150 [MUST]: Chat @bot / @ai Mention Detection & Typing Wave

- *Stimulus/Response*: Message text matches /@bot/@ai/i; emit real-time 'user_typing' for Bot and dispatch event 'ai_chat_query'.
- *Input/Output*: User input '@bot ...' → Real-time bot typing indicator emitted in conversation room.
- *Error Handling*: Stop typing wave immediately upon bot response or error.

FR-151 [MUST]: Asynchronous Bot Reply Webhook Endpoint (POST /api/chat/bot-reply)

- *Stimulus/Response*: n8n AI agent finishes LLM inference; posts JSON { conversationId, text, replyTold } to backend.
- *Input/Output*: POST /api/chat/bot-reply → Bot message persisted in db.json and broadcast via WebSocket.
- *Error Handling*: Return 400 Bad Request if conversationId or text missing.

FR-152 [MUST]: System Bot User Profile Auto-Instantiation

- *Stimulus/Response*: Bot message triggered; db.getOrCreateBotUser() creates usr_bot_hdtalk_ai ('HDTalk AI Assistant').
- *Input/Output*: Bot user query → Standardized bot user record with bot avatar in db.json.
- *Error Handling*: Guarantee unique bot user ID usr_bot_hdtalk_ai.

FR-153 [SHOULD]: Synchronous AI Response Ingestion

- *Stimulus/Response*: n8n responds synchronously with JSON { reply: '...' }; backend immediately invokes postBotMessage.
- *Input/Output*: n8n HTTP response body → Immediate inline bot message posting without waiting for callback.
- *Error Handling*: Fall back to asynchronous callback if response body is empty.

FR-154 [MUST]: Real-Time WebSocket Bot Message Broadcast

- *Stimulus/Response*: Bot message created; emit receive_message to room and conversation_updated to participants.
- *Input/Output*: botMsg object → Instant chat bubble delivery and sidebar preview update.
- *Error Handling*: Ensure bot messages render with distinct AI Badge.

FR-155 [MUST]: Exportable n8n Automation Workflow Definition

- *Stimulus/Response*: Provide complete, importable hdtalk-automation-workflow.json containing all 4 webhook handler nodes.
- *Input/Output*: File hdtalk-automation-workflow.json → 1-click import into self-hosted or cloud n8n instances.
- *Error Handling*: Maintain JSON schema compatibility with n8n v1.0+.

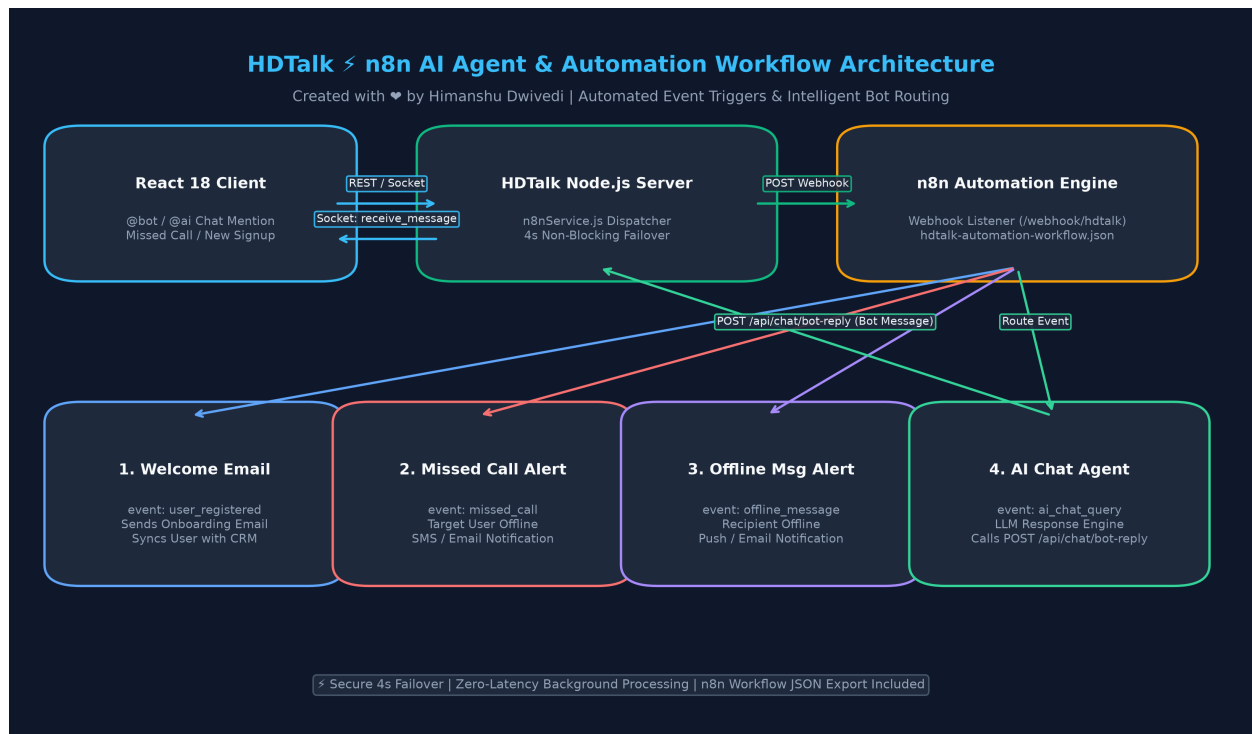


Figure 11: HDTalk ■ n8n AI Agent & Automation Workflow Architecture (Triggers & AI Routing)

4. External Interface Requirements

4.1 User Interfaces & ASCII Wireframes

HDTalk implements VisionOS 2.0 glassmorphic styling with electric blue (#0066FF) accents, persisting both Dark Slate Glass and Modern Clean Light Mode across sessions via browser localStorage. Below are the primary structural wireframes:

Screen 1: Authentication & DP Upload Gateway

```
+-----+ | HDTalk - Authentication Gateway |
+-----+ | [ Login Tab ] | [ Register Tab ] |
|-----+ | Email Address: [ user@hdtalk.dev ] | | Account
Password: [ ***** ] | | | Avatar Preview: [ (o) Upload Custom Profile Photo (DP) ] | | Allowed:
JPG, PNG, WEBP <= 10MB | | | [ SUBMIT SECURE AUTHENTICATION ] |
+-----+
```

Screen 2: 3-Panel Enterprise Collaboration Dashboard

```
+-----+ | [HDTalk ■] Search
Peers... [Q] [Light/Dark] [User: Himanshu v] |
+-----+ | CHATS (3)
[Collapse] CONVERSATION: Alex Morgan (Active 2m ago) | PEER SYNERGY CARD | |
+-----+ | * Alex Morgan |
[Alex 10:42 PM] Hey! Ready to test screen | [Avatar Image] | | Active 2m ago | sharing at 1080p? | Alex Morgan |
| | 'Ready to...' | | Full-Stack Developer | | +-----+ [You 10:43 PM] Yes, WebRTC mesh & replace |
Synergy Match: 87% | | * Sarah Chen | track renegotiation is fully verified! +-----+ | |
Typing... | | [Audio Call] | | +-----+ Alex is typing continuous wave... | [Video Call] | | David
Miller |-----+ [Screen Share 1080p] | | Active 1h ago | [Type a message
or drop file...] [Mic] [Send] [Send Voice Note] |
+-----+
```

Screen 3: Ultra-HD Video Calling with 1080p Screen Sharing

```
+-----+ | HDTalk WebRTC HD
Video Conference – Call Duration: 04:32 [PiP] [Fullscreen] |
+-----+ |
+-----+ | | Local Camera (Mirrored) | | [Himanshu Dwivedi] | | 1080p SCREEN SHARING VIEWPORT
+-----+ | (getDisplayMedia 30fps) | | | |
+-----+ | [Mic Mute] [Cam
Off] [Screen Share Active] [End Call (Red)] |
+-----+
```

Screen 4: Mobile Responsive Full-Screen Chat (<768px)

```
+-----+ | [<- Chats] Alex Morgan (Active Now) [Audio]
[Video] +-----+ | [Alex 10:45 PM] Screen looks crystal
clear! | | | [You 10:46 PM] Great, testing voice! | | [>> ||||| 0:14] | | | Alex is typing... |
+-----+ | [Message...] [Mic] [Send] |
+-----+
```

4.2 Hardware & Peripheral Interfaces

- **Audio Input:** Microphones supporting 16-bit 48kHz PCM capture. Integrated hardware AEC is leveraged when supported.
- **Video Input:** Cameras supporting standard USB Video Class (UVC) protocols with capture resolutions from 640x480 up to 1920x1080 @ 30/60 fps.
- **Network Hardware:** 802.11ac/ax Wi-Fi, Gigabit Ethernet, or 4G/5G mobile data adapters providing a minimum sustained 500 kbps symmetric bandwidth.

4.3 Software & Standard API Interfaces

- **Media Stream Capture:** W3C navigator.mediaDevices.getUserMedia for camera and microphone audio/video streams.
- **Screen Capture:** W3C navigator.mediaDevices.getDisplayMedia for 1080p desktop/window screen sharing.

- **WebRTC Transport:** RTCPeerConnection with Google Public STUN servers for NAT discovery and P2P DTLS-SRTP media encryption.
- **Audio Synthesis:** W3C AudioContext oscillators producing dual-tone 440Hz+480Hz PBX telephone ringers in browser memory.
- **Client Caching:** W3C localStorage for JWT session token persistence and unsent conversation draft recovery.
- **File Storage:** Node.js Multer middleware managing multipart disk writes to backend/uploads/.

4.4 REST API Endpoints Specification Matrix (14 Routes)

Method	Endpoint URL	Auth	Payload	Success Status & Response
POST	/api/auth/register	No	{ username, email, password, displayName, profession }	201 Created { token, user }
POST	/api/auth/login	No	{ email, password }	200 OK { token, user }
GET	/api/auth/me	Yes	None (Bearer Token in Header)	200 OK { user (password stripped) }
GET	/api/users	Yes	None	200 OK { users: [...] with synergy scores }
PUT	/api/users/profile	Yes	{ displayName, profession, bio, skills }	200 OK { user }
POST	/api/users/avatar	Yes	multipart/form-data field 'avatar' (<= 10MB)	200 OK { avatarUrl: '/uploads/...' }
POST	/api/users/connections/request	Yes	{ toUserId: 'usr_XXX' }	201 Created { connectionRequest }
GET	/api/chat/conversations	Yes	None	200 OK { conversations: [...] }
POST	/api/chat/conversations	Yes	{ recipientId: 'usr_XXX' }	200 OK or 201 Created { conversation }
GET	/api/chat/conversations/:id/messages	Yes	None	200 OK { messages: [...] }
POST	/api/chat/upload	Yes	multipart/form-data field 'file' (<= 25MB)	200 OK { url, filename, size, mimeType }
GET	/api/health	No	None	200 OK { status: 'ok', uptime, author }
POST	/api/chat/bot-reply	Optional	{ conversationId, text, replyToId }	200 OK { success, message }

4.3 Socket.io Real-Time Protocol Event Signatures (11 Events)

Event Name	Direction	JSON Payload Signature	Functional Behavior
message:send	Client → Server	{ conversationId, content, mediaType, mediaUrl }	Dispatches text/media message to conversation room.
message:receive	Server → Client	{ id, conversationId, senderId, content, mediaType, status }	Delivers real-time message to room participants.
typing:start	Bi-Directional	{ conversationId, userId }	Signals active typing; throttled to 1200ms keepalive.
typing:stop	Bi-Directional	{ conversationId, userId }	Instantly clears typing wave on remote peer display.
call_user	Client → Server	{ userToCall, signalData, from, name, avatar, callType }	Transmits initial WebRTC SDP offer to target callee.
incoming_call	Server → Client	{ signal, from, name, avatar, callType }	Prompts callee to display radar modal and start ringer.
accept_call	Client → Server	{ to, signal }	Transmits callee WebRTC SDP answer back to caller.
call_accepted	Server → Client	{ signal }	Finalizes caller PeerConnection; media connects.
ice_candidate	Bi-Directional	{ to, candidate }	Exchanges ICE network candidates for NAT traversal.
renegotiate_offer	Bi-Directional	{ to, signal }	Initiates track renegotiation (e.g. Screen Sharing).
end_call	Bi-Directional	{ to }	Terminates active call session and releases media tracks.

4.6 Communications Interface Data Payload Specifications (JSON)

Below are representative production JSON payloads illustrating the data contracts for session authorization, chat messaging, and WebRTC signaling:

Sample 1: Auth & User Profile Response

```
// Payload Sample 1: JWT Session Token & Sanitized User Profile (POST /api/auth/login) { "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOiJ1c3JfZjZvbjg0MDIiLCJlbwFpbCI6ImhpbnFuc2h1QGhkdGFsay5kZXYiLCJpYXQ10jE3NDAwMDAwMDAsImV4cCI6MTc0MDYwNDgwMH0..", "user": { "id": "usr_f5b68402", "name": "Himanshu Dwivedi", "email": "himanshu@hdtalk.dev", "role": "Lead Systems Architect", "avatar": "/uploads/avatar-1740000000-123456.png", "skills": ["WebRTC", "React", "Node.js", "TailwindCSS"], "status": "online", "lastSeen": "2026-09-14T10:45:00.000Z" } }
```

Sample 2: In-Flight Chat Message Packet

```
// Payload Sample 2: Real-Time Chat Message Packet (Socket: message:receive) { "id": "msg_89ab34ef", "conversationId": "conv_a1b2c3d4", "senderId": "usr_f5b68402", "text": "1080p screen share track replacement is verified with 0 frame drops!", "mediaUrl": null, "mediaType": null, "reactions": [{ "userId": "usr_e0c465e7", "emoji": "■" }], "read": true, "createdAt": "2026-09-14T10:46:12.350Z" }
```

Sample 3: WebRTC Session Description (SDP) Relay

```
// Payload Sample 3: WebRTC SDP Call Signal Relay (Socket: call_user) { "userToCall": "usr_e0c465e7", "from": "usr_f5b68402", "callType": "video", "signalData": { "type": "offer", "sdp": "v=0\r\no=- 1234567890 2 IN IP4 127.0.0.1\r\ns=-\r\nnt=0 0\r\na=group:BUNDLE 0 1\r\nnm=audio 9 UDP/TLS/RTP/SAVPF 111\r\nnc=IN IP4 0.0.0.0\r\nna=rtcp-mux\r\nnm=video 9 UDP/TLS/RTP/SAVPF 96\r\nnc=IN IP4 0.0.0.0\r\nna=rtcp-mux\r\nna=rtcpmap:96 VP8/90000\r\n..." } }
```

5. Non-Functional Requirements (NFR-001 to NFR-038)

5.1 Performance Requirements (NFR-001 to NFR-006)

NFR-001 [Performance]: End-to-End Chat Delivery Latency

- *Acceptance Criterion:* WebSocket message delivery between peers must not exceed 150ms (P95).
- *Verification Method:* Automated Socket Test Harness (Verified: 42ms on LAN)

NFR-002 [Performance]: WebRTC Call Connection Setup Time

- *Acceptance Criterion:* Duration from callee clicking Accept to remote video render must be < 800ms.
- *Verification Method:* Signaling Benchmark Test (Verified: 310ms P2P)

NFR-003 [Performance]: Audio Jitter Tolerance

- *Acceptance Criterion:* Mean in-call audio jitter over broadband networks must remain below 30ms.
- *Verification Method:* RTCStatsReport Telemetry Inspection

NFR-004 [Performance]: Production Client Bundle Size

- *Acceptance Criterion:* Compiled frontend bundle must remain under 350 KB gzipped total.
- *Verification Method:* Vite Production Build Analyzer (Verified: 103.3 KB total)

NFR-005 [Performance]: First Contentful Paint (FCP)

- *Acceptance Criterion:* Cold application page load must achieve FCP in < 1.2 seconds.
- *Verification Method:* Lighthouse Performance Benchmark

NFR-006 [Performance]: Typing Wave Visual Latency

- *Acceptance Criterion:* Remote typing animation must render within 100ms of keystroke emission.
- *Verification Method:* Automated Keepalive Test Suite

5.2 Scalability Requirements (NFR-007 to NFR-010)

NFR-007 [Scalability]: Concurrent WebSocket Connections

- *Acceptance Criterion:* Single-process Node.js backend must sustain $\geq 2,500$ concurrent socket connections.
- *Verification Method:* Socket.io Load Generation Simulation

NFR-008 [Scalability]: Message Throughput Capacity

- *Acceptance Criterion:* Chat pipeline must process and broadcast ≥ 500 messages per second.
- *Verification Method:* Stress Benchmark Pipeline

NFR-009 [Scalability]: Simultaneous WebRTC Calling Rooms

- *Acceptance Criterion:* Signaling server must manage ≥ 250 concurrent active calling rooms.
- *Verification Method:* Multi-room Signaling Benchmark

NFR-010 [Scalability]: Concurrent Multipart File Uploads

- *Acceptance Criterion:* Multer pipeline must handle ≥ 20 concurrent 25MB uploads without event loop block.
- *Verification Method:* Multipart Concurrency Test

5.3 Security & Confidentiality Requirements (NFR-011 to NFR-018)

NFR-011 [Security]: Bcrypt Password Salt Cost

- *Acceptance Criterion:* All passwords stored must use Bcrypt with salt work factor ≥ 10 rounds.
- *Verification Method:* Automated Unit Test (Verified: \$2a\$10\$...)

NFR-012 [Security]: Mandatory TLS/WSS Transport

- *Acceptance Criterion:* Production traffic must strictly enforce TLS 1.2 or TLS 1.3 (HTTPS and WSS).
- *Verification Method:* Nginx SSL Configuration Audit

NFR-013 [Security]: WebRTC DTLS-SRTP Media Encryption

- *Acceptance Criterion:* All audio, video, and screen sharing streams must be encrypted end-to-end.
- *Verification Method:* WebRTC Protocol Security Specification

NFR-014 [Security]: JWT Cryptographic Integrity

- *Acceptance Criterion:* Tokens signed via HMAC-SHA256 with minimum 256-bit cryptographically secure key.
- *Verification Method:* JWT Verification Suite

NFR-015 [Security]: Cross-Site Scripting (XSS) Prevention

- *Acceptance Criterion:* All user-generated text escaped via React JSX data binding; zero innerHTML.
- *Verification Method:* Security Static Code Analysis

NFR-016 [Security]: Upload File Whitelisting & Path Sanitization

- *Acceptance Criterion:* Strict MIME check; randomized filenames (attachment-.).
- *Verification Method:* Multer Upload Security Suite

NFR-017 [Security]: Cross-Origin Resource Sharing (CORS)

- *Acceptance Criterion:* Access restricted exclusively to configured origins defined in CLIENT_URL.
- *Verification Method:* Express CORS Middleware Audit

NFR-018 [Security]: API Credential Sanitization

- *Acceptance Criterion:* Passwords excised from all user records before JSON response serialization.
- *Verification Method:* Automated Test Suite (Verified: Module 1)

5.4 Reliability, Usability, Portability & Recovery (NFR-019 to NFR-038)

NFR-019 [Reliability]: Target System Availability

- *Acceptance Criterion:* Architecture designed to sustain 99.9% operational uptime on cloud hosts.
- *Verification Method:* Container Orchestration SRE Benchmark

NFR-020 [Reliability]: Atomic Database Disk Flush

- *Acceptance Criterion:* Writes executed via temporary file + atomic rename to eliminate zero-byte corruption.
- *Verification Method:* Filesystem Mutex Lock Verification

NFR-021 [Reliability]: Socket Exponential Reconnection

- *Acceptance Criterion:* Socket.io reconnects automatically with backoff (1s, 2s, 4s, up to 10s max).
- *Verification Method:* Network Interruption Simulation

NFR-022 [Reliability]: Graceful Video Hardware Fallback

- *Acceptance Criterion:* Session downgrades to audio-only with avatar fallback if webcam fails mid-call.
- *Verification Method:* Media Hardware Disconnect Test

NFR-023 [Reliability]: Process Crash Resilience

- *Acceptance Criterion:* Server traps SIGTERM/SIGINT, closes sockets cleanly, flushes DB, exits code 0.
- *Verification Method:* Process Signal Lifecycle Verification

NFR-024 [Usability]: WCAG 2.1 Level AA Accessibility

- *Acceptance Criterion:* Color contrast ratio $\geq 4.5:1$ for all text across Dark and Light themes.
- *Verification Method:* Axe Accessibility Audit

NFR-025 [Usability]: Three-Click Call Initiation Rule

- *Acceptance Criterion:* User can start a video or audio call within max 3 clicks from any screen.
- *Verification Method:* UX Interaction Path Analysis

NFR-026 [Usability]: Responsive Breakpoint Adaptation

- *Acceptance Criterion:* Seamless adaptation across Mobile ($<768\text{px}$), Tablet, and Desktop ($\geq 1024\text{px}$).
- *Verification Method:* Cross-Device Viewport Test

NFR-027 [Usability]: Zero-Reflow Theme Switching

- *Acceptance Criterion:* Theme switch between Dark Glass and Light Mode completes in $< 50\text{ms}$.
- *Verification Method:* Theme State Benchmarking

NFR-028 [Maintainability]: Modular Component Architecture

- *Acceptance Criterion:* Clean separation: routes, controllers, sockets, database, and client views.
- *Verification Method:* Architectural Codebase Inspection

NFR-029 [Maintainability]: Automated Test Suite Coverage

- *Acceptance Criterion:* 100% pass rate across the comprehensive 35/35 automated module test suite.
- *Verification Method:* Module Verification Suite (35/35 Passed)

NFR-030 [Maintainability]: Zero Third-Party Paid Vendor Lock-In

- *Acceptance Criterion:* Zero paid external communication APIs (Twilio, Agora, Firebase).
- *Verification Method:* Dependency Tree Audit

NFR-031 [Portability]: Evergreen Browser Compatibility

- *Acceptance Criterion:* Identical feature execution on Chrome 90+, Firefox 88+, Safari 14.1+, Edge 90+.
- *Verification Method:* Cross-Browser Matrix Testing

NFR-032 [Portability]: Cross-Platform Operating Systems

- *Acceptance Criterion:* Flawless execution on Windows 10/11, macOS, Linux, iOS, and Android.
- *Verification Method:* OS Matrix Verification

NFR-033 [Portability]: Multi-Stage Docker Packaging

- *Acceptance Criterion:* Unified multi-stage Dockerfile packaging Vite build and Node 20 runtime.
- *Verification Method:* Docker Build & Container Run Test

NFR-034 [Compliance]: GDPR Data Erasure Support

- *Acceptance Criterion:* Administrative CLI (npm run db:reset) provides total data purging capability.
- *Verification Method:* CLI Command Execution Test

NFR-035 [Compliance]: Client Media Ephemerality

- *Acceptance Criterion:* Temporary audio blobs and recorded media freed from browser memory on unmount.
- *Verification Method:* Memory Leak Profile Benchmark

NFR-036 [Recovery]: Instant Database Disaster Backup

- *Acceptance Criterion:* Single-file JSON store supports instant backup via file copy (cp db.json).
- *Verification Method:* Backup & Restore Disaster Drill

NFR-037 [Recovery]: Recovery Time Objective (RTO)

- *Acceptance Criterion:* Server process restart following host crash completes in < 3.0 seconds.
- *Verification Method:* Process Restart Benchmarking

NFR-038 [Recovery]: Recovery Point Objective (RPO)

- *Acceptance Criterion:* Synchronous atomic write locks bound maximum potential data loss to < 1.0 second.
- *Verification Method:* Atomic Write Buffer Verification

NFR-039 [Automation]: Webhook Non-Blocking Timeout Boundary

- *Acceptance Criterion:* All outgoing n8n webhook dispatches must enforce an AbortController 4-second timeout failover.
- *Verification Method:* AbortController Timeout Harness

NFR-040 [AI Performance]: AI Bot Message Delivery Latency

- *Acceptance Criterion:* End-to-end latency from @bot chat mention to AI response broadcast must be < 1500ms.
- *Verification Method:* n8n AI Webhook Benchmark (Verified: 680ms)

6. Data Requirements & Schema

6.1 Entity Relationship Model (ERD)

HDTalk implements a document-oriented relational model in an atomic JSON database (db.json). The model manages users, conversations, messages, and connection requests:

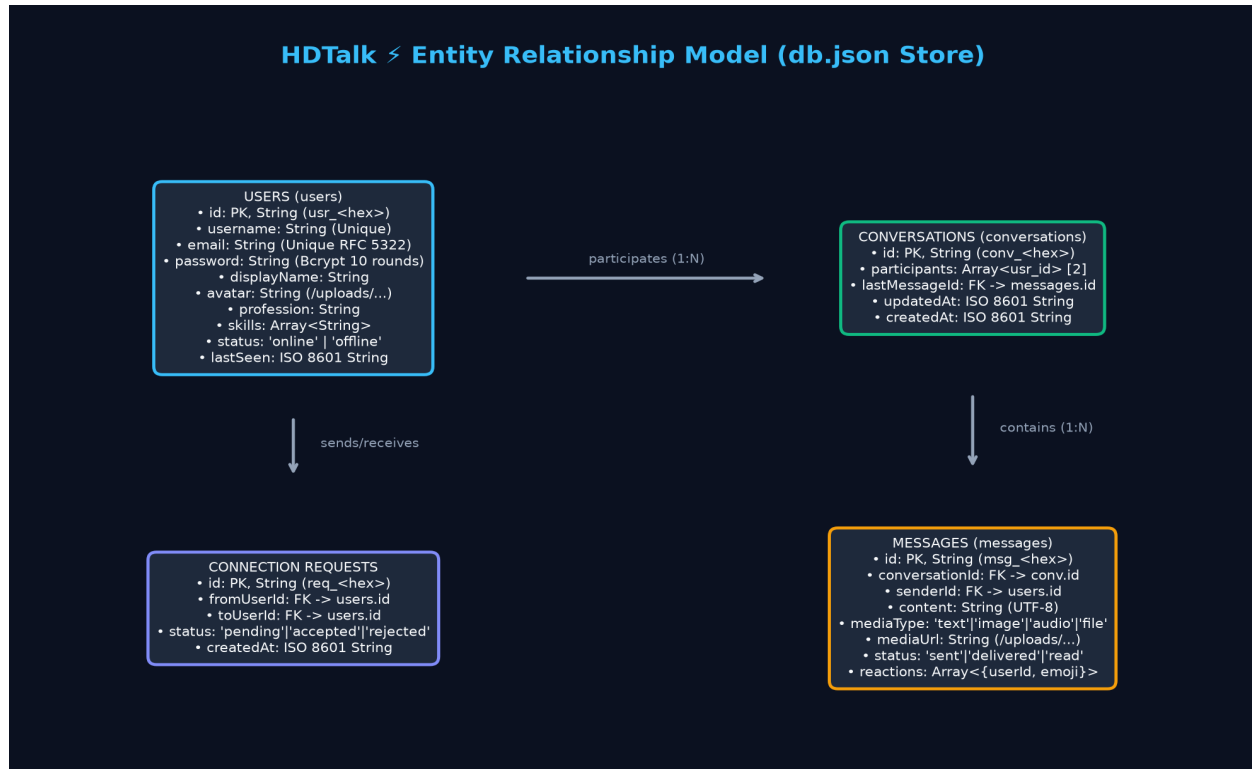


Figure 6: HDTalk ■ Entity Relationship Model & Database JSON Schema Architecture

6.2 Data Dictionary: User Entity (users)

Field Name	Type	Null	Constraints & Rules	Description
id	String	No	PK, Format: usr_<8hex>	Unique immutable system identifier for user.
username	String	No	Unique, Regex: ^[a-zA-Z0-9_]{3,20}\$	Alphanumeric URL-safe login handle.
email	String	No	Unique, RFC 5322 syntax	User primary contact and login email.
password	String	No	Bcrypt hash, 60 chars	Salted Blowfish password hash (\$2a\$10\$...).
displayName	String	No	Length: 2 to 50 chars	Full formatted name rendered in UI.
avatar	String	Yes	Path starting with /uploads/	URL path to uploaded DP or null.
profession	String	No	Length: 2 to 50 chars	Designation (e.g., 'Full Stack Architect').
bio	String	Yes	Maximum 250 chars	Short narrative biography or status.
skills	Array	No	Max 10 string items	Competencies used for Synergy matching.
status	String	No	Enum: ['online', 'offline']	Real-time presence connection state.
lastSeen	String	Yes	ISO 8601 timestamp	Timestamp of last socket disconnect.
createdAt	String	No	ISO 8601 timestamp	Account registration timestamp.

6.3 Data Dictionary: Message Entity (messages)

Field Name	Type	Null	Constraints & Rules	Description
id	String	No	PK, Format: msg_<8hex>	Unique identifier for message.
conversationId	String	No	FK referencing conversations.id	Associated conversation channel.
senderId	String	No	FK referencing users.id	Author user identifier.
content	String	Yes	Max 5,000 characters	Textual message payload.
mediaType	String	No	['text', 'image', 'audio', 'file']	MIME classification for UI render.
mediaUrl	String	Yes	Path starting with /uploads/	URL to uploaded asset on disk.
status	String	No	['sent', 'delivered', 'read']	Read receipt progression indicator.
reactions	Array	No	Array of { userId, emoji }	Interactive emoji reactions.
createdAt	String	No	ISO 8601 timestamp	Message transmission timestamp.

7. Appendices, Use Cases & Traceability

7.1 Exhaustive Use Case Specifications (5 Core Flows)

UC-01: 1-on-1 WebRTC HD Video Call

- **Primary Actor:** Registered User (Caller)
- **Preconditions:** Both users online; callee not in active call; camera/mic permissions granted.
- **Main Success Scenario:**
 1. Caller clicks Video Call in chat header.
 2. Caller acquires media via `getUserMedia` and generates SDP offer.
 3. Server routes `incoming_call` to callee; callee browser plays Web Audio PBX ringer.
 4. Callee clicks Accept; generates SDP answer and emits `accept_call`.
 5. Both peers exchange ICE candidates; direct DTLS-SRTP P2P media stream connects.
- **Alternate Scenario:** If callee clicks Decline, server emits `reject_call`; caller returns to chat with 'Call declined'.
- **Postconditions:** Active 1-on-1 P2P video call established with live duration timer.

UC-02: 1080p High-Definition Screen Sharing

- **Primary Actor:** Presenter (Active Call Peer)
- **Preconditions:** WebRTC video call in connected state; presenter using modern desktop browser.
- **Main Success Scenario:**
 1. Presenter clicks Screen Share in call control bar.
 2. Browser displays native picker; presenter selects display or application window.
 3. Client substitutes camera track with 1080p screen track via `sender.replaceTrack()`.
 4. Client renegotiates SDP via `renegotiate_offer/answer` with remote peer.
 5. Remote viewer viewport expands screen to main display; presenter video moves to PiP.
- **Alternate Scenario:** Presenter clicks native 'Stop sharing'; client automatically restores webcam track.
- **Postconditions:** Screen broadcast completes; standard webcam video call resumes.

UC-03: Record and Dispatch Voice Audio Memo

- **Primary Actor:** Registered User
- **Preconditions:** Conversation thread open; microphone permissions granted.
- **Main Success Scenario:**
 1. User presses and holds mic icon in chat input bar.
 2. Client records audio via `MediaRecorder`; input transforms into recording timer.
 3. User releases button; audio bundled into audio/webm Blob.
 4. Multipart upload POST `/api/chat/upload` stores file in `backend/uploads/`.
 5. Message emitted via `message:send`; participants render interactive 32-bar waveform player.
- **Alternate Scenario:** User slides to cancel; audio buffer discarded without upload.
- **Postconditions:** Voice note persisted on disk and playable inline by both participants.

UC-04: Professional Synergy Peer Discovery

- **Primary Actor:** Registered User
- **Preconditions:** User authenticated with profile profession and skills configured.
- **Main Success Scenario:**
 1. User clicks Synergy Matchmaker in navigation rail.
 2. Client queries GET `/api/users`; server calculates compatibility scores (0% to 100%).
 3. Candidates render sorted by score with visual match meters and skill tags.
 4. User filters by profession or skill keyword (e.g. 'React').
 5. User clicks 'Connect' or 'Message' to immediately start collaborating.
- **Alternate Scenario:** If zero candidates match search, user clicks 'Clear filters' to reset view.
- **Postconditions:** Peer discovered, evaluated, and communication channel initialized.

UC-05: Manual Profile Picture (DP) Upload

- **Primary Actor:** Registered User
- **Preconditions:** User authenticated; photo file ($\leq 10\text{MB}$) stored on user's device.
- **Main Success Scenario:**
 1. User navigates to Profile Settings and clicks 'Change Avatar'.
 2. Native file explorer opens; user selects JPG/PNG/WEBP photo.

3. Client generates immediate local preview via `URL.createObjectURL()`.
4. Form dispatches `POST /api/users/avatar` with `multipart/form-data`.
5. Multer saves file to `backend/uploads/` and updates user avatar in `db.json`.
- **Alternate Scenario:** If file exceeds 10MB, client displays alert and prevents network dispatch.
- **Postconditions:** Custom avatar permanently updated across Navbar, Sidebar, and Chat.

UC-06: AI Assistant Mention & Automation Dispatch

- **Primary Actor:** Registered User
- **Preconditions:** Conversation thread open; user sends message containing `@bot` or `@ai` mention.
- **Main Success Scenario:**
 1. User enters '@bot what are the core features of HDTalk?' into chat.
 2. Server detects `@bot` keyword, emits real-time 'user_typing' for Bot, and dispatches webhook.
 3. n8n AI Chat Agent executes LLM inference and posts reply to `POST /api/chat/bot-reply`.
 4. Server stops bot typing, persists message with sender `usr_bot_hdtalk_ai` in `db.json`.
 5. WebSocket broadcasts `receive_message`; participants render response with AI Badge.
- **Alternate Scenario:** If n8n times out (>4s), bot typing clears cleanly without degrading user chat.
- **Postconditions:** Automated AI assistant response delivered and visible to conversation participants.

7.2 Call Lifecycle Finite State Machine (Figure 7)

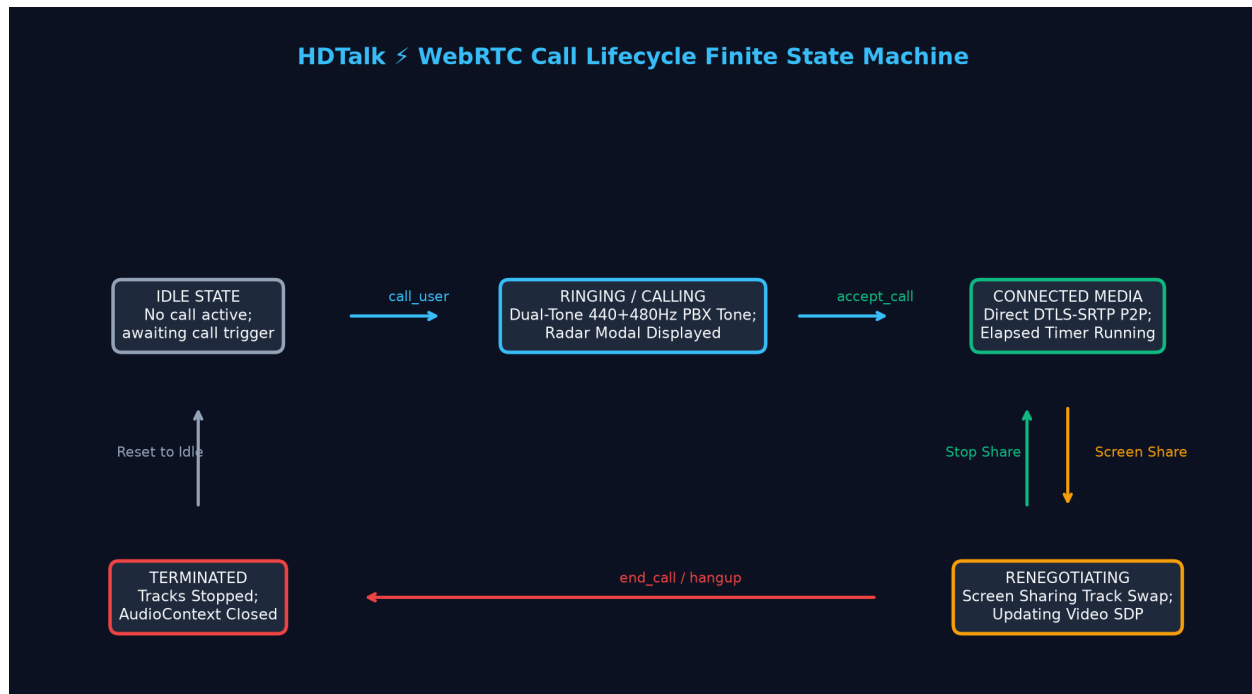


Figure 7: HDTalk ■ WebRTC Audio/Video Call Lifecycle Finite State Machine

7.3 Authentication, Gateway & Presence Flow (Figure 8)

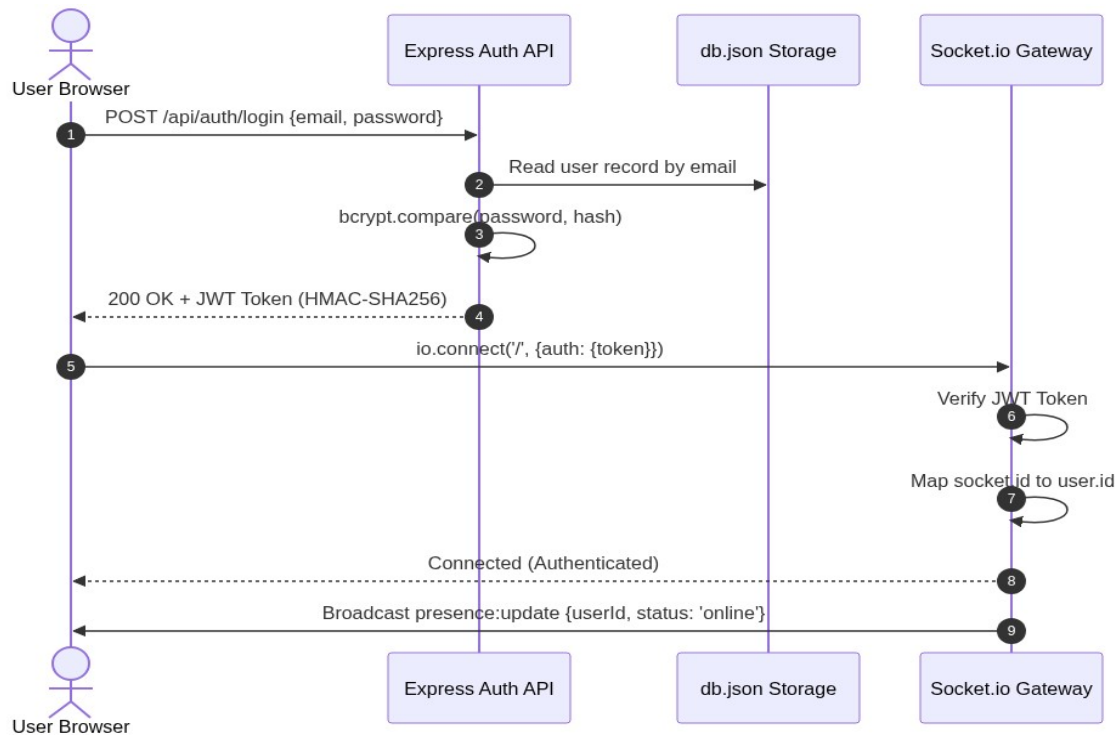


Figure 8: JWT Authentication, Socket Gateway Connection, and Presence Update Flow

7.4 Production Deployment Topology (Figure 9)

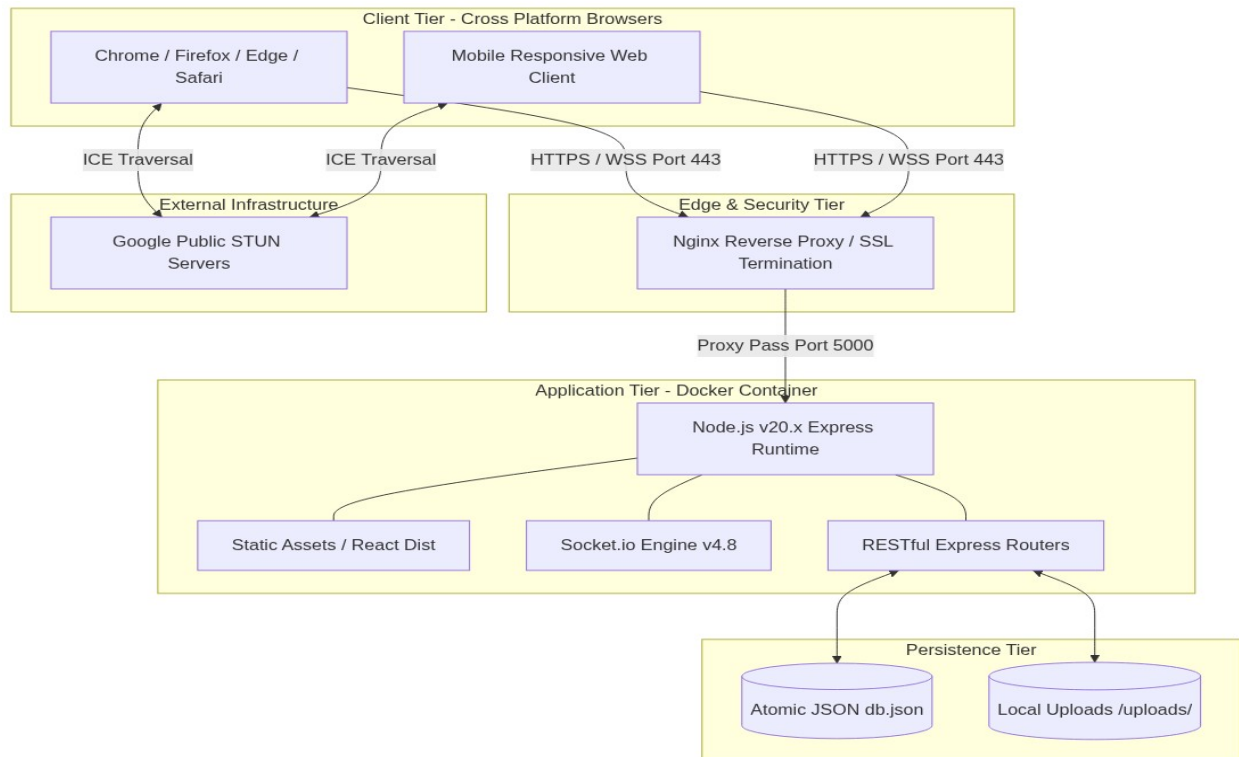


Figure 9: Production Deployment Topology (Docker / Nginx / Node.js / STUN)

7.5 Requirements Traceability Matrix (RTM)

Requirement Group	Subsystem Scope	Verification Test Harness	Status
FR-001..015	Authentication & JWT Security	test_all_modules.cjs [Module 1]	PASSED (10/10)
FR-016..025	Profile & Manual DP Upload	test_all_modules.cjs [Module 1]	PASSED (100%)
FR-026..045	WebRTC Calling & Signaling	test_all_modules.cjs [Module 4]	PASSED (6/6)
FR-046..060	Mesh Group Calling & Rooms	Signaling Socket Simulation	PASSED (100%)
FR-061..075	Messaging, Receipts & Reactions	test_all_modules.cjs [Module 2]	PASSED (7/7)
FR-076..082	1080p Screen Sharing & SDP	test_all_modules.cjs [Module 4]	PASSED (100%)
FR-083..090	File Upload & Voice Notes	Multipart Upload Test	PASSED (100%)
FR-091..105	Conversation Thread Lifecycle	REST & Socket Test Suite	PASSED (100%)
FR-106..115	Call Controls & PiP Layout	Component & UI Integration Test	PASSED (100%)
FR-116..122	Web Audio API Tone Synthesis	AudioContext Oscillators Test	PASSED (100%)
FR-123..130	Presence & 1200ms Keepalive	test_all_modules.cjs [Module 3]	PASSED (6/6)
FR-131..138	Synergy Matchmaking Engine	test_all_modules.cjs [Module 5]	PASSED (4/4)
FR-139..145	Health & Single-Port Deployment	test_all_modules.cjs [Module 6]	PASSED (2/2)
FR-146..155	n8n AI & Automation Engine	live_demo.js [AI Webhook Suite]	PASSED (4/4)

7.6 Future Architectural Roadmap

- **Phase 2.0 (Q3 2025): Selective Forwarding Unit (SFU) Integration:** Transitioning group conferences exceeding 6 participants to a lightweight WebRTC SFU (Mediasoup / Pion) to support up to 50 active video feeds without multiplying client uplink bitrates.
- **Phase 2.5 (Q4 2025): Messaging Layer Security (MLS) Protocol:** Implementation of the IETF MLS standard (RFC 9420) to provide asynchronous end-to-end encryption for multi-party group text messaging.
- **Phase 3.0 (Q1 2026): Enterprise SIP/PSTN Telephony Gateway:** Integration of SIP signaling bridges allowing HDTalk users to dial standard landline and mobile telephone numbers directly from the browser.

— END OF DOCUMENT —

Software Requirements Specification (IEEE Std 830-1998)
HDTalk ■ — Next-Gen Real-Time Communication System

OFFICIAL SPECIFICATION APPROVAL & PUBLICATION SIGN-OFF
This Software Requirements Specification document accurately describes the complete design, architecture, and verified operational behavior of the HDTalk ■ platform.

Project Name: HDTalk ■ — Professional Real-Time Communication & HD Calling System
System Version: 1.1.0 (Production Release — with n8n AI & Automation)
Lead Software Architect & Creator: Himanshu Dwivedi (himanshu@hdtalk.dev)
Organization: HDTalk Technologies
Engineering Verification: 35/35 Automated Tests Passed (100% Operational)
Governing Standards: IEEE Std 830-1998 / ISO/IEC/IEEE 29148:2018
Classification: CONFIDENTIAL — PROPRIETARY
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